# All two-$\rho$ terms proportional to $\log(\vee/\Lambda)$, and how JIMWLK and BFKL come out of them

genuine terms, terms from reordering three $\rho$'s, terms from reordering four $\rho$'s

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# 1 Question and answer

**Question.** All three-$\rho$ terms proportional to $\log(\vee/\Lambda)$ cancel. Give all the two-$\rho$ terms proportional to $\log(\vee/\Lambda)$:

- those from reordering the three-$\rho$ terms;
- the genuine two-$\rho$ terms of `Evolution_two_rho.tex`;
- those from reordering the four-$\rho$ terms.

Show how they give the BFKL and the JIMWLK evolution. You also object to the statement that BFKL comes only with $\log(\vee/k^+)$: BFKL and JIMWLK are both proportional to $\log(\vee/\Lambda)$. No limits are put on $\vee$, $p^+$, $k^+$, $\Lambda$; all that is known is $\vee > p^+, k^+ > \Lambda$.

**Answer.**

1. **All two-$\rho$ terms proportional to** $\log(\vee/\Lambda)$ are listed in Secs. 5–7, source by source and term by term, in common variables:

   - from reordering the three-$\rho$ terms (Sec. 5: 10 rows of Groups I–III, and $\mathcal{D}$);
   - the genuine two-$\rho$ terms (Sec. 6: Group I Rows II, V; Group II Rows I, IV; Group III Row III part 2);
   - from reordering the four-$\rho$ terms (Sec. 7: the lists of `4rho.tex`, and the two-$\rho$ terms of the three-$\rho$ remainders A1–B2).

   There are 2386 two-$\rho$ terms before collecting. After the colour algebra and collecting equal terms within each source, 440 entries remain (Table 2).

2. **Their sum is JIMWLK⊗LO exactly** (Sec. 8). This holds in all 13 kernel structures and all 154 independent colour structures, with exact coefficients (polynomials in $N_c$). It holds pointwise in the position $z$ of the unobserved gluon. Nothing is left over.

3. **Why** $\log(\vee/\Lambda)$ **multiplies JIMWLK, with no limit taken** (Sec. 3). Every logarithm comes from a $p^+$ integral, and each integral is treated exactly.

   - $\Lambda$ enters a row only through those limits of its $p^+$ integral at which a gluon carries plus momentum $\Lambda$ ($p^+ = \Lambda$, $k^+ - p^+ = \Lambda$ or $|p^+ - k^+| = \Lambda$). That gluon is softer than the measured gluon.
   - $\vee$ enters only through the upper limits $p^+ = \vee$ or $p^+ + k^+ = \vee$. There the unobserved gluon is harder than the measured gluon.
   - Of your two logarithms, $\log(\vee/\Lambda)$ and $\log(\vee/k^+)$, only the first contains $\Lambda$. So its coefficient is fixed entirely by the $\Lambda$ end of the integrals, i.e. by the gluon softer than the measured one.

   Integrating out a gluon softer than the measured gluon is one step of the evolution of the target: JIMWLK. A row is an exact identity of the form $\mathfrak{T}\log(\vee/\Lambda) + (\mathfrak{P} - \mathfrak{T})\log(\vee/k^+) + R$, with $\mathfrak{T}$ ($\mathfrak{P}$) the integrand at the $\Lambda$ ($\vee$) end and $R$ free of both logarithms. The coefficients are unique (Sec. 3.2).

4. **Where BFKL is.** BFKL contains terms in which the measured gluon is emitted by the unobserved gluon at $z$, e.g. $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,K(x,x',z)$. Such a kernel occurs only at the $\vee$ end, where the unobserved gluon is the harder one. Sec. 3.3 shows this on your Group II, Row IV:

   - the same $p^+$ integral contains your $\log(\vee/\Lambda)$ term of `Evolution_two_rho.tex` (the $\Lambda$ end, JIMWLK);
   - it also contains the BFKL term $2N_c\,K(w,w',z)K(x,x',z)\,[U(z)-U(x')][U(z)-U(x)]\rho\rho$ (the $\vee$ end).

   The BFKL term comes with $\log\vee$ and without $\log(1/\Lambda)$; in your variables that is $\log(\vee/k^+)$. It is therefore in the coefficient of $\log(\vee/k^+)$ of that row, and never in the coefficient of $\log(\vee/\Lambda)$. No kernel of this kind occurs in any $\log(\vee/\Lambda)$ term of any source (Table 4).

5. **How BFKL comes out** (Sec. 9). For every row, add the coefficient of $\log(\vee/k^+)$ to that of $\log(\vee/\Lambda)$. This gives the coefficient of $\log\vee$, $P_{\text{row}}$:

   - $P_{\text{row}} = A_{\text{row}}$ for the two-$\rho$ terms from the four-$\rho$ terms;
   - $P_{\text{row}} = 0$ for Group I Row V, $\mathcal{D}$ and the genuine Group I terms;
   - otherwise $P_{\text{row}}$ is the integrand at the $\vee$ end.

   The sum is BFKL⊗LO exactly: 9 kernel structures, 114 independent colour structures, exact coefficients.

6. **The result.** With $\mathcal{N}_0 = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}$, $L_\Lambda = \log(\vee/\Lambda)$, $L_k = \log(\vee/k^+)$, and $\mathcal{J}$, $\mathcal{B}$ the JIMWLK and BFKL evolution of the LO cross section per unit rapidity (Secs. 8, 9), the logarithms are treated exactly:

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\rm LL} = \mathcal{N}_0\Big[L_\Lambda\,\mathcal{J} + L_k\,(\mathcal{B}-\mathcal{J})\Big] = \mathcal{N}_0\Big[\log\frac{k^+}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,\mathcal{B}\Big].} \tag{1.1}$$

- The target evolves over the rapidity interval below the measured gluon, $\log(k^+/\Lambda)$. The projectile evolves over the interval above it, $\log(\vee/k^+)$. Together these make $\delta Y = \log(\vee/\Lambda)$.
- In your variables, $\log(\vee/\Lambda)$ multiplies $\mathcal{J}$ alone.
- The same result can be written so that both evolutions multiply $\delta Y$. The identity is

$$\left.\frac{d^3N}{d^3k}\right|_{\rm LL} = \mathcal{N}_0\Big[\tfrac{1}{2}\log\frac{\vee}{\Lambda}\,(\mathcal{J}+\mathcal{B}) + \tfrac{1}{2}\log\frac{k^{+2}}{\vee\Lambda}\,(\mathcal{J}-\mathcal{B})\Big]. \tag{1.2}$$

  The second term vanishes only when the measured gluon sits in the middle of the window, $k^+ = \sqrt{\vee\Lambda}$. Otherwise it is there, because $\mathcal{J} \neq \mathcal{B}$ (Table 5).

7. **Fixes used** (all found in the earlier notes; Table 1). Groups II and III are correct as written, except F5.

| fix | what | $\log(\vee/\Lambda)$ | $\log\vee$ |
|---|---|---|---|
| F1 | `4rho.tex`, first part: drop the $-8N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,K(y,x',z)$ term | ✓ | ✓ |
| F2 | `4rho.tex`, $dN^{(1)}$ term 2: repeated index; correct term $3f^{abc}f^{deg}\rho^{b'}(x')\rho^c(y)U^{ag}(y)U^{db'}(w')U^{eb}(z)$ | ✓ | ✓ |
| F3 | Group I Row III: part 2 (edge kernel $\Phi$) $\to \mathcal{D}$ | ✓ | – |
| F4 | Group I Row II, $U(w)\mathbb{A}^{(3)}$: $\mathfrak{F}_\alpha$ goes with $-\log(\vee/k^+)$, not $\log(k^+/\Lambda)$ ($p^+$ ordering of the virtual factor) | ✓ | ✓ |
| F5 | Group III Row III part 2 (genuine two-$\rho$): overall sign | ✓ | ✓ |
| F6 | Group I Row II (genuine two-$\rho$): bracket $\to \frac{1}{\pi}\int_z[K(x,w,z)-K(w,w,z)]$ (pole and constant only) | ✓ | – |
| P1 | Group I Row III, $p^+ > k^+ + \Lambda$: large-$p^+$ limit $-2K_h \to +2K_h$ | – | ✓ |

Table 1: The fixes, and which coefficient they change.

Everything uses your convention $U^{\dagger bc} = U^{bc} = U^{cb}$.

Table 2: The sources of two-$\rho$ terms proportional to $\log(\vee/\Lambda)$. "Terms": ordered two-$\rho$ products produced by the reordering (each kernel monomial and the c.c. separately). "Entries": distinct (kernel, colour structure) pairs after the colour algebra and collecting, as listed in Secs. 5–7.

| label | source | terms | entries |
|---|---|---|---|
| $\mathrm{I.II}^{3\rho}$ | Group I, Row II | 720 | 22 |
| $\mathrm{I.III}^{3\rho}$ | Group I, Row III (part 1) | 72 | 21 |
| $\mathcal{D}$ | $\mathcal{D}$, the replacement of Group I, Row III, part 2 | 72 | 24 |
| $\mathrm{I.IV}^{3\rho}$ | Group I, Row IV | 144 | 46 |
| $\mathrm{I.V}^{3\rho}$ | Group I, Row V | 288 | 16 |
| $\mathrm{II.I}^{3\rho}$ | Group II, Row I | 288 | 31 |
| $\mathrm{III.I}^{3\rho}$ | Group III, Row I | 144 | 46 |
| $\mathrm{III.II}^{3\rho}$ | Group III, Row II | 144 | 45 |
| $\mathrm{III.III}^{3\rho}$ | Group III, Row III (part 1) | 144 | 16 |
| $\mathrm{III.V}^{3\rho}$ | Group III, Row V | 72 | 24 |
| $\mathrm{III.VI}^{3\rho}$ | Group III, Row VI | 72 | 24 |
| $\mathrm{I.II}^{\rm gen}$ | Group I, Row II | 16 | 16 |
| $\mathrm{I.V}^{\rm gen}$ | Group I, Row V | 64 | 16 |
| $\mathrm{II.I}^{\rm gen}$ | Group II, Row I | 16 | 16 |
| $\mathrm{II.IV}^{\rm gen}$ | Group II, Row IV | 12 | 12 |
| $\mathrm{III.III}^{\rm gen}$ | Group III, Row III (part 2) | 16 | 15 |
| A,B | Remainders A1, A2, B1, B2 | 72 | 20 |
| $4\rho$ | The lists of `4rho.tex` | 30 | 30 |

## 2 Conventions

**Units and logarithms.**

$$\mathcal{N}_0 \equiv \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}, \qquad L_\Lambda \equiv \log\frac{\vee}{\Lambda}, \qquad L_k \equiv \log\frac{\vee}{k^+}, \qquad \mathcal{N} \equiv \mathcal{N}_0\,L_\Lambda. \tag{2.1}$$

So $\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda}=2\mathcal{N}$, $\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\log\frac{\vee}{\Lambda}=\pi N_c\mathcal{N}$, $\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\log\frac{\vee}{\Lambda}=2\pi\mathcal{N}$. A coefficient written as a number is in units of $\mathcal{N}$. Your rules are

$$\log\frac{\Lambda}{k^+}=-L_\Lambda+L_k,\qquad \log\frac{\vee-k^+}{\Lambda}=L_\Lambda+\log\Big(1-\frac{k^+}{\vee}\Big). \tag{2.2}$$

For every row we write the large logarithms as

$$\frac{d^3N_{\rm row}}{d^3k}=\mathcal{N}_0\Big[A_{\rm row}\,L_\Lambda+B_{\rm row}\,L_k\Big]+\big(\text{no }L_\Lambda,\ \text{no }L_k\big),\qquad P_{\rm row}\equiv A_{\rm row}+B_{\rm row}. \tag{2.3}$$

$A_{\rm row}$ is what `Evolution_two_rho.tex`, `Evolution_three_rho.tex` and `4rho.tex` give (blue logarithms). Since both $L_\Lambda$ and $L_k$ contain $\log\vee$, $P_{\rm row}$ is the coefficient of $\log\vee$.

**Kernels.** $\mathcal{K}(X)\equiv X/X^2$ (a two-vector), $\mathcal{K}(-X)=-\mathcal{K}(X)$, $\mathcal{K}(X)\cdot\mathcal{K}(X)=1/X^2$, and

$$K(a,b,z)\equiv\mathcal{K}(a-z)\cdot\mathcal{K}(b-z)=\frac{(a-z)\cdot(b-z)}{(a-z)^2(b-z)^2},\qquad K(a,a,z)=\frac{1}{(a-z)^2}. \tag{2.4}$$

**Common variables.** All transverse positions except $w$, $w'$ are integrated, and the phase $e^{-ik\cdot(w'-w)}$ is always present. The measured gluon sits at $w$ in the amplitude and at $w'$ in the conjugate amplitude; the unobserved gluon sits at $z$. The two charges are called $x$ and $x'$:

- the charge tied to $w$ by its emission kernel is $x$, and the one tied to $w'$ is $x'$;
- if one charge is tied to both, it is called $x'$ and the other one $x$.

Every term is written as a coefficient times [kernel] times [colour structure]. A kernel structure is a product of two factors $\mathcal{K}\cdot\mathcal{K}$. We label the structures of the $\log(\vee/\Lambda)$ terms $S_1,\dots,S_{13}$ and those of the $\log\vee$ terms $S'_1,\dots,S'_9$ (Table 4).

**Charges.** The valence charges obey $[\rho^a(x),\rho^b(y)]=if^{abc}\rho^c(x)\,\delta^{(2)}(x-y)$. After reordering, the two charges of a two-$\rho$ term form a symmetric product and behave as commuting variables. A two-$\rho$ term has a real coefficient, so its complex conjugate is the same term with $w\leftrightarrow w'$.

**Colour.** Colour structures are simplified with $U^{bc}=U^{cb}$, $U^{ab}(p)U^{ac}(p)=\delta^{bc}$, $f^{abc}U^{ad}(p)U^{be}(p)U^{cg}(p)=f^{deg}$ and $f^{acd}f^{bcd}=N_c\delta^{ab}$. Dummy indices are renamed canonically. Equal colour structures (up to sign) are collected. Linear relations between different colour structures (Jacobi-type identities) are used only in the comparison tables (Secs. 8, 9); they are listed in App. A.

# 3 Which logarithm multiplies which evolution, with no limits

## 3.1 Where $\Lambda$ and $\vee$ enter

Every row of the notes is an integral over the plus momentum $p^+$ of the unobserved (real or virtual) gluon. The limits are built from $\Lambda$, $k^+$ and $\vee$:

- Groups II and III (real gluon, $p^++k^+\le\vee$): $\Lambda<p^+<\vee-k^+$;
- Group I, Row V (the measured gluon splits into $p^+$ and $k^+-p^+$): $\Lambda<p^+<k^+-\Lambda$;
- Group I, Row III: $\Lambda<p^+<k^+-\Lambda$ and $k^++\Lambda<p^+<\vee$;
- virtual terms: $\Lambda<p^+<\vee$, split at $k^+$ where the integrand changes form.

Once the $p^+$ integral is done, a row is an exact function of $\Lambda$, $k^+$, $\vee$ (and of transverse positions). $\Lambda$ and $\vee$ appear in it only through the limits.

- A limit containing $\Lambda$ is a point where one gluon carries plus momentum $\Lambda$: the unobserved gluon ($p^+=\Lambda$), or the partner of the measured gluon in a splitting ($k^+-p^+=\Lambda$, $|p^+-k^+|=\Lambda$). That gluon is softer than the measured gluon.
- A limit containing $\vee$ is $p^+=\vee$ or $p^++k^+=\vee$: the unobserved gluon is harder than the measured gluon.

No relation between $\Lambda$, $k^+$ and $\vee$ beyond $\vee>p^+,k^+>\Lambda$ is used.

## 3.2 An exact identity

Take a row with integrand $I(p^+)$ over $\Lambda < p^+ < \vee - k^+$ (transverse positions fixed). Write $p^+ = u\,k^+$ and

$$G(u) \equiv p^+\, I(p^+)\Big|_{p^+=uk^+}, \qquad \mathfrak{T} \equiv G(0), \qquad \mathfrak{P} \equiv G(\infty). \tag{3.1}$$

$\mathfrak{T}$ and $\mathfrak{P}$ are properties of the integrand (its behaviour at the two ends), not of $\Lambda$ or $\vee$. Splitting the range at $u = 1$ and adding and subtracting $\mathfrak{T}$ below and $\mathfrak{P}$ above,

$$\begin{aligned}\int_\Lambda^{\vee-k^+} dp^+\, I(p^+) &= \int_{\Lambda/k^+}^{\vee/k^+-1} \frac{du}{u}\, G(u) \\ &= \mathfrak{T}\,\log\frac{k^+}{\Lambda} + \mathfrak{P}\,\log\frac{\vee - k^+}{k^+} + \int_{\Lambda/k^+}^{1} \frac{du}{u}\big[G(u) - \mathfrak{T}\big] + \int_1^{\vee/k^+-1} \frac{du}{u}\big[G(u) - \mathfrak{P}\big].\end{aligned} \tag{3.2}$$

This is exact for every $\Lambda < k^+ < \vee - \Lambda$. (If $\vee < 2k^+$ the last integral runs backwards; the identity is unchanged.) With your rules, $\log(k^+/\Lambda) = L_\Lambda - L_k$ and $\log\frac{\vee-k^+}{k^+} = L_k + \log(1 - k^+/\vee)$:

$$\boxed{\int_\Lambda^{\vee-k^+} dp^+\, I(p^+) = \mathfrak{T}\,L_\Lambda + \big(\mathfrak{P} - \mathfrak{T}\big)\,L_k + R,} \qquad R = \mathfrak{P}\log\Big(1 - \frac{k^+}{\vee}\Big) + \int_{\Lambda/k^+}^{1} \frac{du}{u}\big[G - \mathfrak{T}\big] + \int_1^{\vee/k^+-1} \frac{du}{u}\big[G - \mathfrak{P}\big]. \tag{3.3}$$

$R$ contains neither $L_\Lambda$ nor $L_k$. The integrands of its two integrals vanish at $u \to 0$ and $u \to \infty$ respectively, so the integrals are finite for every value of $\Lambda/k^+$ and $k^+/\vee$ between 0 and 1, the end points included. $\log(1 - k^+/\vee)$ is the function your second rule already puts aside.

**The coefficients are unique.** Suppose the same row is also written as $A'L_\Lambda + B'L_k + R'$ with $R'$ of the same kind. Then

$$(\mathfrak{T} - A')\log\frac{k^+}{\Lambda} + (\mathfrak{P} - A' - B')\log\frac{\vee}{k^+} = R' - R \qquad \text{for all } \Lambda < k^+ < \vee. \tag{3.4}$$

Making $\Lambda$ smaller at fixed $k^+$, $\vee$ makes the left side arbitrarily large unless $A' = \mathfrak{T}$. Making $\vee$ larger at fixed $\Lambda$, $k^+$ gives $A' + B' = \mathfrak{P}$. So nothing is chosen. This uses only that the identity holds for all values; no approximation is made.

**Other ranges.**

- $\Lambda < p^+ < \vee$: the same, with $\log(\vee/k^+)$ in place of $\log\frac{\vee-k^+}{k^+}$.
- $\Lambda < p^+ < k^+ - \Lambda$: both limits contain $\Lambda$. Let $\mathfrak{T}$ and $\bar{\mathfrak{T}}$ be the integrand at $p^+ \to 0$ and at $k^+ - p^+ \to 0$. Then the row is $(\mathfrak{T} + \bar{\mathfrak{T}})\log(k^+/\Lambda) + R$, so

  $$A = \mathfrak{T} + \bar{\mathfrak{T}}, \qquad B = -A, \qquad P = 0. \tag{3.5}$$

  There is no $\vee$, and so no contribution to the evolution of the projectile (Group I Row V, the genuine Group I terms, $\mathcal{D}$).
- A pole at $p^+ = k^+$ cut off at $k^+ \pm \Lambda$ (the edge logarithms of Group I Row III) is again a $\Lambda$ end.

**What this means.** For every row,

$$A_{\text{row}} = \text{the integrand at the } \Lambda \text{ ends}, \qquad P_{\text{row}} = A_{\text{row}} + B_{\text{row}} = \text{the integrand at the } \vee \text{ end}. \tag{3.6}$$

- At a $\Lambda$ end one gluon is softer than the measured gluon. It is emitted by everything harder (the charges and the measured gluon) and scatters on the target. Integrating it out is one step of the evolution of the target: JIMWLK.
- At the $\vee$ end the unobserved gluon is harder than the measured one. It belongs to the projectile, and the measured gluon can be emitted by it. Integrating it out is one step of the evolution of the projectile: BFKL.

Hence

$$\sum_{\text{rows}} A_{\text{row}} = \mathcal{J} \quad (\text{JIMWLK} \otimes \text{LO}), \qquad \sum_{\text{rows}} P_{\text{row}} = \mathcal{B} \quad (\text{BFKL} \otimes \text{LO}), \qquad \sum_{\text{rows}} B_{\text{row}} = \mathcal{B} - \mathcal{J}. \tag{3.7}$$

These are expectations. Secs. 8 and 9 check both sums term by term and do not use this argument.

### 3.3 Example: your Group II, Row IV

In this row a gluon of momentum $p^+ + k^+$ is emitted by the charges at $x$, $x'$ and splits into the measured gluon ($k^+$, at $w$) and the unobserved gluon ($p^+$, at $z$). The parent gluon sits at the momentum-weighted position

$$y = \frac{k^+ w + p^+ z}{k^+ + p^+} = w + \frac{u}{1+u}\,(z-w), \qquad u = \frac{p^+}{k^+}, \qquad \Lambda < p^+ < \vee - k^+. \tag{3.8}$$

- $\Lambda$ *end,* $u \to 0$*:* $y \to w$. The parent is the measured gluon, and the gluon at $z$ is a soft gluon emitted by it in the amplitude and in the conjugate amplitude:

$$\mathfrak{T}_{\text{II.IV}} = 2N_c\, K(w,w',z)\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\left[U^{e'c}(w') - U^{e'c}(x')\right]\left[U^{ec}(w) - U^{ec}(x)\right]\rho^{e'}(x')\rho^{e}(x). \tag{3.9}$$

  Integrated over $z$ in dimensional regularization, $\int_z K(w,w',z) = \pi[\frac{1}{\bar\epsilon} - \log(4\pi^2\mu^2(w-w')^2)]$, this is exactly the term of `Evolution_two_rho.tex` (blue $\log(\vee/\Lambda)$). Its kernel $K(w,w',z)$ (soft gluon attached to the measured gluon) is a JIMWLK kernel: structure $S_6$.

- $\vee$ *end,* $u \to \infty$*:* $y \to z$. The parent is the gluon at $z$, emitted by the charges, and it emits the measured gluon:

$$\mathfrak{P}_{\text{II.IV}} = 2N_c\, K(w,w',z)\, K(x,x',z)\left[U^{e'c}(z) - U^{e'c}(x')\right]\left[U^{ec}(z) - U^{ec}(x)\right]\rho^{e'}(x')\rho^{e}(x). \tag{3.10}$$

  This is the BFKL term in which both measured gluons come from $z$ (second line of Eq. (9.1)): structure $S_5'$.

By Eq. (3.3) the row is exactly

$$\mathcal{N}_0\Big[\mathfrak{T}_{\text{II.IV}}\,\log\frac{\vee}{\Lambda} + \big(\mathfrak{P}_{\text{II.IV}} - \mathfrak{T}_{\text{II.IV}}\big)\log\frac{\vee}{k^+}\Big] + R_{\text{II.IV}}. \tag{3.11}$$

- In your `G2` derivation the first term is the blue $\log(\vee/\Lambda)$.
- The second sits inside the remaining integral over $\xi = k^+/(k^+ + p^+)$ from $k^+/\vee$ to 1: its end $\xi = k^+/\vee$ produces the $\log(\vee/k^+)$.
- The BFKL term $\mathfrak{P}_{\text{II.IV}}$ multiplies $\log(\vee/k^+) = \log\vee - \log k^+$: it comes with $\log\vee$ and without $\log(1/\Lambda)$.

This is the general pattern. A kernel in which the measured gluon is emitted by the gluon at $z$ needs the gluon at $z$ to be the harder one, and that happens only at the $\vee$ end. So no such kernel multiplies $\log(1/\Lambda)$. In your variables $\log\vee$ sits in both $L_\Lambda$ and $L_k$, while $\log(1/\Lambda)$ sits only in $L_\Lambda$. A term that comes with $\log\vee$ but not with $\log(1/\Lambda)$ is a multiple of $L_k$. So a BFKL kernel appears in $B_{\text{row}}$ and in $P_{\text{row}} = A_{\text{row}} + B_{\text{row}}$, and never in $A_{\text{row}}$.

### 3.4 Row by row at a test point

Table 3 gives $A_{\text{row}}$, $P_{\text{row}}$ and $B_{\text{row}} = P_{\text{row}} - A_{\text{row}}$ for every source at one random configuration (positions, symmetric SU(3) Wilson lines, charges). Each value is the integrand of Eq. (2.3) at fixed $z$ (all kernels and colour factors evaluated, no integration).

| source | $A_{\text{row}}$ (coeff. of $L_\Lambda$) | $P_{\text{row}}$ (coeff. of $\log\vee$) | $B_{\text{row}}$ (coeff. of $L_k$) | |
|---|---|---|---|---|
| I.II$^{3\rho}$ | 0.017076 | 0.009958 | −0.007118 | |
| I.III$^{3\rho}$ | 0.024871 | −0.030301 | −0.055172 | |
| $\mathcal{D}$ | −0.034889 | 0.000000 | 0.034889 | P=0 |
| I.IV$^{3\rho}$ | 0.014443 | −0.006278 | −0.020721 | |
| I.V$^{3\rho}$ | 0.028152 | 0.000000 | −0.028152 | P=0 |
| II.I$^{3\rho}$ | −0.086062 | −0.098052 | −0.011990 | |
| III.I$^{3\rho}$ | −0.014443 | 0.006278 | 0.020721 | |
| III.II$^{3\rho}$ | 0.061023 | 0.057347 | −0.003676 | |
| III.III$^{3\rho}$ | −0.057722 | 0.022818 | 0.080540 | |
| III.V$^{3\rho}$ | 0.034889 | 0.005430 | −0.029459 | |
| III.VI$^{3\rho}$ | −0.049755 | 0.000910 | 0.050665 | |
| I.II$^{\text{gen}}$ | −0.022099 | 0.000000 | 0.022099 | P=0 |
| I.V$^{\text{gen}}$ | −0.136196 | 0.000000 | 0.136196 | P=0 |
| II.I$^{\text{gen}}$ | −0.055693 | −0.056134 | −0.000442 | |
| II.IV$^{\text{gen}}$ | −0.005613 | 0.044201 | 0.049814 | |
| III.III$^{\text{gen}}$ | 0.129956 | 0.060542 | −0.069414 | |
| A,B | −0.046743 | −0.046743 | 0.000000 | P=A |
| $4\rho$ | 0.018846 | 0.018846 | 0.000000 | P=A |
| sum | −0.179957 | −0.011178 | 0.168778 | |
| target | $\mathcal{J} = -0.179957$ | $\mathcal{B} = -0.011178$ | $\mathcal{B} - \mathcal{J} = 0.168778$ | |

Table 3: The two-$\rho$ content of each source at one test point (seed 3). The rows with $P = A$ are the two-$\rho$ terms from the four-$\rho$ terms: there the integrand is $1/p^+$ times a $p^+$-independent factor over the whole window. The rows with $P = 0$ have both limits at $\Lambda$ ends.

# 4 The reordering

An ordered product of three charges $A$, $B$, $C$ (each $\rho^a(x)$ at its own position) is written as its fully symmetrized part plus two-$\rho$ terms. With $S(ABC) = \frac{1}{6}(ABC + ACB + BAC + BCA + CAB + CBA)$:

$$ABC = S(ABC) + \tfrac{1}{4}\Big(\big\{[A,B],C\big\} + \big\{[A,C],B\big\} + \big\{[B,C],A\big\}\Big) + (\text{one } \rho), \tag{4.1}$$

$$A\{B,C\} = 2S(ABC) + \tfrac{1}{2}\big\{[A,B],C\big\} + \tfrac{1}{2}\big\{[A,C],B\big\} + (\text{one } \rho), \tag{4.2}$$

$$\{B,C\}A = 2S(ABC) - \tfrac{1}{2}\big\{[A,B],C\big\} - \tfrac{1}{2}\big\{[A,C],B\big\} + (\text{one } \rho). \tag{4.3}$$

Each commutator gives $if\rho\,\delta^{(2)}$, so each anticommutator $\{[A,B],C\}$ is a two-$\rho$ term with the positions of $A$ and $B$ set equal. The symmetrized three-$\rho$ parts cancel (three-$\rho$ note). The one-$\rho$ terms vanish for a colour-neutral projectile. The derivation and every intermediate step (commutators, $\delta$-functions, colour algebra before and after) are in `ThreeRho_to_TwoRho.pdf` and in the appendices of the combined note. Here we give, for every source, the ordered input and the resulting two-$\rho$ terms.

The four-$\rho$ terms are reordered in the same way, Sec. 7. Their two-$\rho$ terms come in two pieces: those obtained directly (the lists of `4rho.tex`), and those of the three-$\rho$ remainders A1–B2 (one more reordering step).

**How to read the entry tables.** Each row of a table is one entry:

**S:** the kernel structure $S_n$ (Table 4);

**kernel:** the kernel in common variables;

**coeff.:** the coefficient in units of $\mathcal{N}$;

**colour structure:** with the two charges as a symmetric (commuting) product.

The term is

$$\mathcal{N}\int e^{-ik\cdot(w'-w)}\int_{x,x',z}(\text{coeff.})\times(\text{kernel})\times(\text{colour structure}). \tag{4.4}$$

The c.c. of Groups I and III is included. Within a source, entries with the same kernel and the same colour structure (up to sign) are added.

# 5 Two-$\rho$ terms from reordering the three-$\rho$ terms

The inputs are the three-$\rho$ terms of `Evolution_three_rho.tex`, in the form used in the three-$\rho$ note (prefactor in units of $\mathcal{N}$, kernel, sub-terms with their charge order). Labels: I.II$^{3\rho}$ = two-$\rho$ terms from the three-$\rho$ terms of Group I, Row II, etc.

## 5.1 Group I, Row II (I.II$^{3\rho}$)

*Input.* In the notes,

$$\begin{aligned}\frac{d^3N_2}{d^3k} =& -\frac{1}{(2\pi)^3}\frac{g^4 f^{abc}}{64\pi^6}\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2\mathbf{p}\,d^2\mathbf{k}\int_{x,y} e^{i\mathbf{k}\cdot(w-y)}e^{i\mathbf{p}\cdot(y-x)}\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,B^i(\mathbf{k},\mathbf{p})\,\log\frac{\vee}{\Lambda}\\ &\times\Big[U^{ab'}(x')\rho^{b'}(x') - U^{ab'}(w')\rho^{b'}(x')\Big]\Big[U^{ab}(w)\{\rho^b(x),\rho^c(y)\} - U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\}\Big] + \text{c.c.},\\ B^i(\mathbf{k},\mathbf{p}) \equiv& -\frac{\mathbf{k}^2\mathbf{p}^i - (\mathbf{k}\cdot\mathbf{p})\mathbf{k}^i}{(\mathbf{k}-\mathbf{p})^2} - 2\mathbf{k}^i - \frac{\mathbf{p}^2\mathbf{k}^i + \mathbf{k}^2\mathbf{p}^i}{(\mathbf{k}-\mathbf{p})^2}.\end{aligned}$$

Two remarks before reordering.

1. *Colour.* As written, the indices $a$ and $b$ each appear three times in the first term of the second bracket: in $f^{abc}$, $U^{ab'}(x')$, $U^{ab}(w)$ and $\rho^b(x)$. This term is $U^{ab}(w)\mathbb{A}^{(3)b}$ with $\mathbb{A}^{(3)b} \propto f^{bcd}\{\rho^c(x),\rho^d(y)\}$, so we read it as $U^{ah}(w)f^{hbc}\{\rho^b(x),\rho^c(y)\}$. The second term, $f^{abc}U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\} = \bar{\mathbb{A}}^{(3)a}$, is unchanged.

2. *Kernel.* The Fourier transform is done in Part I of the combined note. With $\int d^2\mathbf{p}\,d^2\mathbf{k}\,e^{i\mathbf{k}\cdot(w-y)}e^{i\mathbf{p}\cdot(y-x)}B^i/(\mathbf{p}^2\mathbf{k}^2) = 2\pi i\,\mathfrak{F}^i(w;x,y)$, and since both colour structures are odd under $x\leftrightarrow y$, only the part of $\mathfrak{F}^i$ odd under $x\leftrightarrow y$ contributes. It is

$$\mathfrak{F}^i(w;x,y) = \mathcal{K}^i(x-w)\int_{z'}\big[K(w,y,z') - K(x,y,z')\big] - \mathcal{K}^i(y-w)\int_{z'}\big[K(w,x,z') - K(x,y,z')\big],$$

with $K(a,b,z') = \mathcal{K}(a-z')\cdot\mathcal{K}(b-z')$; equivalently $\mathfrak{F}^i(w;x,y) = -\pi\big[\mathcal{K}^i(x-w)\log\frac{(w-y)^2}{(x-y)^2} - \mathcal{K}^i(y-w)\log\frac{(w-x)^2}{(x-y)^2}\big]$. The prefactor becomes $-\frac{g^4}{64\pi^6}\cdot 2\pi i = -\frac{i}{2}\cdot\frac{g^4}{16\pi^5}$, i.e. $-\frac{i}{2}\mathcal{N}$.

$$\left.\frac{d^3N}{d^3k}\right|_{\rm GI.II} = -\tfrac{1}{2} i\mathcal{N}\int e^{-ik\cdot(w'-w)}\, G_{\rm I.II}(x',x,y)\,\Big\{$$
$$(1)\ +\ U^{ab'}(x')\, f^{hbc}\, U^{ah}(w)\,\rho^{b'}(x')\,\big\{\rho^b(x),\rho^c(y)\big\}$$
$$(2)\ -\ U^{ab'}(x')\, f^{abc}\, U^{bd}(x)\, U^{ce}(y)\,\rho^{b'}(x')\,\big\{\rho^d(x),\rho^e(y)\big\}$$
$$(3)\ -\ U^{ab'}(w')\, f^{hbc}\, U^{ah}(w)\,\rho^{b'}(x')\,\big\{\rho^b(x),\rho^c(y)\big\}$$
$$(4)\ +\ U^{ab'}(w')\, f^{abc}\, U^{bd}(x)\, U^{ce}(y)\,\rho^{b'}(x')\,\big\{\rho^d(x),\rho^e(y)\big\}\Big\} + \text{c.c.} \tag{5.1}$$

with $G_{\rm I.II}(x',x,y) = \mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x,y)$. Reordering: Eq. (4.2) with $A$, $B$, $C$ the charges at $x'$, $x$, $y$. The c.c. is included in the entries below (it is the same term with the phase reversed and the charges in reverse order; in common variables, $w \leftrightarrow w'$). *Fix F4.* In sub-terms 1 and 3 (the $U(w)\mathbb{A}^{(3)}$ term) the kernel $\mathfrak{F}^i$ is replaced by $\mathfrak{F}^i_\beta$: the monomials b and d (the $\mathfrak{F}^i_\alpha$ part) are removed. $\mathfrak{F}^i_\alpha$ is the commutator $[\mathbb{N}^{(2)},\mathbb{A}^{(1)}]$; with the $p^+$ ordering of the virtual factor it multiplies $-\log(\vee/k^+)$ instead of $\log(k^+/\Lambda)$ (combined note, Part I). It reappears in the coefficient of $\log\vee$, Sec. 9.

**Two-$\rho$ terms.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-2N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-1$ | $f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+2N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$ |

## 5.2 Group I, Row III, part 1 (I.III$^{3\rho}$)

*Input.* Part 1 of the notes has three terms with the same colour structure. The first two (red) are equal and opposite and cancel. The third has prefactor $\frac{ig^4 f^{abd}}{16\pi^5} = i\mathcal{N} f^{abd}$ and kernel $K^{(1)} = \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$. Part 2 has five terms, again with the same colour structure and prefactor $\pm\frac{ig^4 f^{abd}}{8\pi^5} = \pm 2i\mathcal{N} f^{abd}$; its terms 1 and 5 are equal, and so are terms 2 and 4. Together,

$$\text{Row III} = i\mathcal{N}\, f^{abd}\int K_{\rm I.III}(x',x,y)\,[\dots][\dots] + \text{c.c.}, \qquad K_{\rm I.III} = K^{(1)} + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi(x),$$
$$\Phi(x) \equiv \frac{1}{(z-w)^2}\left[\frac{2\,(z-w)\cdot(x-z)\,(x-w)^2 - 2\,(z-w)\cdot(x-w)\,(x-z)^2}{\big[(x-w)^2-(x-z)^2\big]^2} - 1\right].$$

$$\left.\frac{d^3N}{d^3k}\right|_{\rm GI.III} = i\mathcal{N}\int e^{-ik\cdot(w'-w)}\, K_{\rm I.III}(x',x,y)\,\Big\{$$
$$(1)\ +\ f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(z)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)$$
$$(2)\ -\ f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)$$
$$(3)\ -\ f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(z)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)$$
$$(4)\ +\ f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)\Big\} + \text{c.c.} \tag{5.2}$$

with $K_{\rm I.III}(x',x,y) = \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z) + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi(x)\Big]$. Reordering: Eq. (4.1) with $A$, $B$, $C$ the charges at $x'$, $x$, $y$. The c.c. is included in the entries below (it is the same term with the phase reversed and the charges in reverse order; in common variables, $w \leftrightarrow w'$). *Fix F3.* Only part 1 (monomial a, the kernel $K^{(1)}$) is kept here. Part 2 (monomial b, the edge kernel $\Phi$) is replaced by $\mathcal{D}$, Sec. 5.3.

**Two-$\rho$ terms.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+N_c$ | $U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x')$ |

## 5.3 $\mathcal{D}$: replacement of Group I, Row III, part 2

This is the term that replaces part 2 of Group I, Row III (fix F3). It was found in the three-$\rho$ cancellation: with it all symmetrized three-$\rho$ terms cancel. Its two-$\rho$ terms follow from the charge order $\rho(x')\rho(x)\rho(y)$ of Row III.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\rm D} \;=\; &-2i\,\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_{\mathcal{D}}(x',x,y)\,\Big\{\\ &(1)\;+\;f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(w)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)\\ &(2)\;-\;f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)\\ &(3)\;-\;f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(w)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)\\ &(4)\;+\;f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)\Big\}+\text{c.c.}\end{aligned}\tag{5.3}$$

with $K_{\mathcal{D}}(x',x,y) = \mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$. Reordering: Eq. (4.1) with $A$, $B$, $C$ the charges at $x'$, $x$, $y$. The c.c. is included in the entries below (it is the same term with the phase reversed and the charges in reverse order; in common variables, $w \leftrightarrow w'$).

**Two-$\rho$ terms.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(w')\,\rho^l(x')\,\rho^h(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(w)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)\,\rho^h(x')$ |

## 5.4 Group I, Row IV (I.IV$^{3\rho}$)

*Input.* Prefactor $\frac{1}{(2\pi)^3}\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda}=2i\mathcal{N}$. The kernel is the sum of the two kernels of the notes, $K_{\mathfrak{B}}\equiv\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\frac{1}{2}\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right]$.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\text{GI.IV}} = 2i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_{\mathfrak{B}}(x',x,y)\,\Big\{\\ (1)\;+\;U^{ab'}(x')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{b'}(x')\,\rho^e(y)\,\rho^{c'}(x)\\ (2)\;-\;U^{ab'}(x')\,f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{b'}(x')\,\rho^e(y)\,\rho^d(x)\\ (3)\;-\;U^{ab'}(w')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{b'}(x')\,\rho^e(y)\,\rho^{c'}(x)\\ (4)\;+\;U^{ab'}(w')\,f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{b'}(x')\,\rho^e(y)\,\rho^d(x)\Big\}+\text{c.c.}\end{aligned}\tag{5.4}$$

with $K_{\mathfrak{B}}(x',x,y)=\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\frac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right]$. Reordering: Eq. (4.1) with $A$, $B$, $C$ the charges at $x'$, $y$, $x$. The c.c. is included in the entries below (it is the same term with the phase reversed and the charges in reverse order; in common variables, $w\leftrightarrow w'$).

**Two-$\rho$ terms.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $+N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |

## 5.5 Group I, Row V ($\mathrm{I.V}^{3\rho}$)

**GI.V1: Group I, Row V, first term (measured gluon at $z$).** *Input.* Prefactor $-\frac{1}{(2\pi)^3}\frac{ig^4}{32\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda}=-\frac{i}{2}\mathcal{N}$. After the $w$ integration with the $\delta$-function of $\mathbb{C}_1$ the measured gluon in the amplitude sits at $z$: the phase is $e^{-ik\cdot(w'-z)}$. The charges sit at $x'$, $y$ and $y'$.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\mathrm{GI.V1}} = &-\tfrac{1}{2}i\mathcal{N}\int e^{-ik\cdot(w'-z)}\,K_{\mathrm{V1}}(x',y,y')\,\Big\{\\ &(1)\;+U^{ab'}(x')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\big\{\rho^e(y),\rho^b(y')\big\}\\ &(2)\;-U^{ab'}(x')\,f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\big\{\rho^d(y),\rho^e(y')\big\}\\ &(3)\;-U^{ab'}(w')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\big\{\rho^e(y),\rho^b(y')\big\}\\ &(4)\;+U^{ab'}(w')\,f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\big\{\rho^d(y),\rho^e(y')\big\}\Big\}+\text{c.c.}\end{aligned}\tag{5.5}$$

with $K_{\mathrm{V1}}(x',y,y')=\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y'-x)$. Reordering: Eq. (4.2) with $A$, $B$, $C$ the charges at $x'$, $y$, $y'$. The c.c. is included in the entries below (it is the same term with the phase reversed and the charges in reverse order; in common variables, $w\leftrightarrow w'$).

**GI.V2: Group I, Row V, second term (measured gluon at $x$).** *Input.* Prefactor $-\frac{i}{2}\mathcal{N}$; the measured gluon in the amplitude sits at $x$ (phase $e^{-ik\cdot(w'-x)}$). Same colour structure as the first term.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\mathrm{GI.V2}} = &-\tfrac{1}{2}i\mathcal{N}\int e^{-ik\cdot(w'-x)}\,K_{\mathrm{V2}}(x',y,y')\,\Big\{\\ &(1)\;+U^{ab'}(x')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\big\{\rho^e(y),\rho^b(y')\big\}\\ &(2)\;-U^{ab'}(x')\,f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\big\{\rho^d(y),\rho^e(y')\big\}\\ &(3)\;-U^{ab'}(w')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\big\{\rho^e(y),\rho^b(y')\big\}\\ &(4)\;+U^{ab'}(w')\,f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\big\{\rho^d(y),\rho^e(y')\big\}\Big\}+\text{c.c.}\end{aligned}\tag{5.6}$$

with $K_{\mathrm{V2}}(x',y,y')=\mathcal{K}(x'-w')\cdot\mathcal{K}(y'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y-z)$. Reordering: Eq. (4.2) with $A$, $B$, $C$ the charges at $x'$, $y$, $y'$. The c.c. is included in the entries below (it is the same term with the phase reversed and the charges in reverse order; in common variables, $w\leftrightarrow w'$).

**Two-$\rho$ terms.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)$ |

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x')$ |

## 5.6 Group II, Row I (II.I$^{3\rho}$)

**GII.I1: Group II, Row I, first term.** *Input.* The prefactor is $-\frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda}=-i\mathcal{N}$ and the kernel is $K_{\mathfrak{B}}$, as in Row IV of Group I. The charges appear as $\{\rho(x'),\rho(y)\}\rho(x)$. Group II has no c.c.; the second term of the row is its conjugate.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\text{GII.I1}} =\; &-i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_{\mathfrak{B}}(x',x,y)\,\Big\{\\ &(1)\;+\;U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,\big\{\rho^{c'}(x'),\rho^b(y)\big\}\,\rho^d(x)\\ &(2)\;-\;U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\big\{\rho^{c'}(x'),\rho^b(y)\big\}\,\rho^e(x)\\ &(3)\;-\;U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,f^{cbd}\,U^{ac}(w)\,\big\{\rho^{d'}(x'),\rho^{e'}(y)\big\}\,\rho^d(x)\\ &(4)\;+\;U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\big\{\rho^{d'}(x'),\rho^{e'}(y)\big\}\,\rho^e(x)\Big\}\end{aligned}\tag{5.7}$$

with $K_{\mathfrak{B}}(x',x,y)=\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\frac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$. Reordering: Eq. (4.3) with $A$, $B$, $C$ the charges at $x$, $x'$, $y$.

**GII.I2: Group II, Row I, second term.** *Input.* Prefactor $+i\mathcal{N}$ and kernel

$$\bar{K}_{\mathfrak{B}}=K_{\mathfrak{B}}\big|_{w\leftrightarrow w'}=\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\,\big[\mathcal{K}(z-w')\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\big].$$

The charges appear as $\rho(x)\{\rho(x'),\rho(y)\}$.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\text{GII.I2}} =\; &i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,\bar{K}_{\mathfrak{B}}(x',x,y)\,\Big\{\\ &(1)\;+\;f^{c'bd'}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x)\,\big\{\rho^c(x'),\rho^b(y)\big\}\\ &(2)\;-\;f^{c'bd'}\,U^{ac'}(w')\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\rho^{d'}(x)\,\big\{\rho^d(x'),\rho^e(y)\big\}\\ &(3)\;-\;f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{ac}(w)\,\rho^{e'}(x)\,\big\{\rho^c(x'),\rho^b(y)\big\}\\ &(4)\;+\;f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\rho^{e'}(x)\,\big\{\rho^d(x'),\rho^e(y)\big\}\Big\}\end{aligned}\tag{5.8}$$

with $\bar{K}_{\mathfrak{B}}(x',x,y)=\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\Big[\mathcal{K}(z-w')\cdot\mathcal{K}(y-z)+\frac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$. Reordering: Eq. (4.2) with $A$, $B$, $C$ the charges at $x$, $x'$, $y$.

**Two-$\rho$ terms.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-N_c$ | $U^{ab}(w)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ce}(x')\,U^{al}(w)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ce}(x')\,U^{ad}(w)\,\rho^{h}(x')\,\rho^{l}(x)$ |

## 5.7 Group III, Row I (III.I$^{3\rho}$)

*Input.* Prefactor $-\frac{1}{(2\pi)^3}\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda}=-2i\mathcal{N}$, kernel $K_{\mathfrak{B}}$. Order of the charges: $\rho(x')\rho(y)\rho(x)$.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{GIII.I}} \;=\; & -2i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_{\mathfrak{B}}(x',x,y)\,\Big\{\\
& (1)\;+\;U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{cbd}\,U^{ac}(w)\,\rho^{b'}(x')\,\rho^{d'}(y)\,\rho^{d}(x)\\
& (2)\;-\;U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{b'}(x')\,\rho^{d'}(y)\,\rho^{e}(x)\\
& (3)\;-\;U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,\rho^{c'}(x')\,\rho^{e'}(y)\,\rho^{d}(x)\\
& (4)\;+\;U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{c'}(x')\,\rho^{e'}(y)\,\rho^{e}(x)\Big\}+\text{c.c.}
\end{aligned}
\tag{5.9}
$$

with $K_{\mathfrak{B}}(x',x,y)=\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\frac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$. Reordering: Eq. (4.1) with $A$, $B$, $C$ the charges at $x'$, $y$, $x$. The c.c. is included in the entries below (it is the same term with the phase reversed and the charges in reverse order; in common variables, $w\leftrightarrow w'$).

**Two-$\rho$ terms.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)$ |

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |

## 5.8 Group III, Row II (III.II$^{3\rho}$)

*Input.* Prefactor $-2i\mathcal{N}$, kernel $K_{\mathfrak{B}}$. Order of the charges: $\rho(y)\rho(x')\rho(x)$.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\text{GIII.II}} = &-2i\,\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_{\mathfrak{B}}(x',x,y)\,\Big\{\\
&(1)\;+\;U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{cbd}\,U^{ac}(w)\,\rho^{d'}(y)\,\rho^{b'}(x')\,\rho^d(x)\\
&(2)\;-\;U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(y)\,\rho^{b'}(x')\,\rho^e(x)\\
&(3)\;-\;U^{ab'}(x')\,f^{cbd}\,U^{ac}(w)\,\rho^b(y)\,\rho^{b'}(x')\,\rho^d(x)\\
&(4)\;+\;U^{ab'}(x')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^b(y)\,\rho^{b'}(x')\,\rho^e(x)\Big\}+\text{c.c.}\end{aligned}\tag{5.10}$$

with $K_{\mathfrak{B}}(x',x,y)=\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\frac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right]$. Reordering: Eq. (4.1) with $A$, $B$, $C$ the charges at $y$, $x'$, $x$. The c.c. is included in the entries below (it is the same term with the phase reversed and the charges in reverse order; in common variables, $w\leftrightarrow w'$).

**Two-$\rho$ terms.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $-1$ | $f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w,z)$ | $-1$ | $f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w,z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $-N_c$ | $U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+1$ | $f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+N_c$ | $U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $-1$ | $f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $+N_c$ | $U^{ab}(z)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)$ |

## 5.9 Group III, Row III, part 1 (III.III$^{3\rho}$)

*Input.* Prefactor $-\frac{1}{(2\pi)^3}\frac{ig^4 f^{c'd'a}}{16\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda} = -i\mathcal{N}f^{c'd'a}$. After the $w'$ integration the measured gluon of the conjugate amplitude sits at $y'$: the phase is $e^{-ik\cdot(y'-w)}$, and the kernel is $K_{\mathfrak{C}} \equiv \mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{GIII.III}} = & -i\mathcal{N}\int e^{-ik\cdot(y'-w)}\,K_{\mathfrak{C}}(x',x,y)\,\Big\{ \\
& (1)\ +\ f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ac}(w)\,\rho^{e'}(x')\,\big\{\rho^c(y),\rho^b(x)\big\} \\
& (2)\ -\ f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\big\{\rho^d(y),\rho^e(x)\big\} \\
& (3)\ -\ f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ac}(w)\,\rho^{e'}(x')\,\big\{\rho^c(y),\rho^b(x)\big\} \\
& (4)\ +\ f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\big\{\rho^d(y),\rho^e(x)\big\}\Big\} + \text{c.c.}
\end{aligned}
\tag{5.11}
$$

with $K_{\mathfrak{C}}(x',x,y) = \mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$. Reordering: Eq. (4.2) with $A$, $B$, $C$ the charges at $x'$, $y$, $x$. The c.c. is included in the entries below (it is the same term with the phase reversed and the charges in reverse order; in common variables, $w \leftrightarrow w'$).

**Two-$\rho$ terms.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ce}(x')\,U^{al}(w)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)\,\rho^h(x')$ |

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ce}(x')\,U^{ad}(w)\,\rho^h(x')\,\rho^l(x)$ |

## 5.10 Group III, Row V (III.V$^{3\rho}$)

*Input.* Prefactor $-\frac{1}{(2\pi)^3}\frac{ig^4 f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda}=-2i\mathcal{N}f^{c'ca}$, phase $e^{-ik\cdot(y'-w)}$, kernel $K_{\mathfrak{C}}$. Order of the charges: $\rho(x')\rho(x)\rho(y)$.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\text{GIII.V}} = &-2i\mathcal{N}\int e^{-ik\cdot(y'-w)}\,K_{\mathfrak{C}}(x',x,y)\,\Big\{\\
&(1)\;+\;f^{c'ca}\,U^{e'c'}(y')\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^e(x)\,\rho^d(y)\\
&(2)\;-\;f^{c'ca}\,U^{e'c'}(y')\,U^{ad}(w)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^e(x)\,\rho^d(y)\\
&(3)\;-\;f^{c'ca}\,U^{c'e'}(x')\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^e(x)\,\rho^d(y)\\
&(4)\;+\;f^{c'ca}\,U^{c'e'}(x')\,U^{ad}(w)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^e(x)\,\rho^d(y)\Big\}+\text{c.c.}\end{aligned}\tag{5.12}$$

with $K_{\mathfrak{C}}(x',x,y)=\mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$. Reordering: Eq. (4.1) with $A$, $B$, $C$ the charges at $x'$, $x$, $y$. The c.c. is included in the entries below (it is the same term with the phase reversed and the charges in reverse order; in common variables, $w\leftrightarrow w'$).

**Two-$\rho$ terms.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(w')\,\rho^l(x')\,\rho^h(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(w)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $+N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)\,\rho^h(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)\,\rho^h(x')$ |

## 5.11 Group III, Row VI (III.VI$^{3\rho}$)

*Input.* Prefactor $-2i\mathcal{N}f^{c'd'a}$, phase $e^{-ik\cdot(y'-w)}$, kernel $K_{\mathfrak{C}}$. Order of the charges: $\rho(x')\rho(y)\rho(x)$.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\text{GIII.VI}} = &-2i\mathcal{N}\int e^{-ik\cdot(y'-w)}\,K_{\mathfrak{C}}(x',x,y)\,\Big\{\\
&(1)\;+\;f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^d(y)\,\rho^e(x)\\
&(2)\;-\;f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^d(y)\,\rho^b(x)\\
&(3)\;-\;f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^d(y)\,\rho^e(x)\end{aligned}$$

$$(4) \;+\; f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{b}(x)\Big\} + \text{c.c.} \tag{5.13}$$

with $K_{\mathfrak{C}}(x',x,y) = \mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$. Reordering: Eq. (4.1) with $A$, $B$, $C$ the charges at $x'$, $y$, $x$. The c.c. is included in the entries below (it is the same term with the phase reversed and the charges in reverse order; in common variables, $w \leftrightarrow w'$).

**Two-$\rho$ terms.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(w)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}f^{deh}\,U^{al}(x')\,U^{bd}(z)\,U^{ce}(w)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(w')\,\rho^{l}(x')\,\rho^{c}(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+1$ | $f^{abc}f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(w)\,\rho^{c}(x')\,\rho^{l}(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-1$ | $f^{abc}f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-1$ | $f^{abc}f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(w')\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $+1$ | $f^{abc}f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{cd}(w')\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $+1$ | $f^{abc}f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(w)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}f^{deh}\,U^{ae}(x')\,U^{bl}(z)\,U^{cd}(w)\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S_{12}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-N_c$ | $U^{ab}(z)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(w')\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}f^{deh}\,U^{ad}(x')\,U^{bl}(z)\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S_{13}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-N_c$ | $U^{ab}(z)\,U^{bc}(x')\,\rho^{c}(x')\,\rho^{a}(x)$ |

# 6 Genuine two-$\rho$ terms (`Evolution_two_rho.tex`)

These are the terms you uploaded. Each is first written in units of $\mathcal{N}$ and in common variables, and then expanded into entries. Group I Rows I, III, IV, Group II Rows II, III and Group III Rows I, II, IV, V, VI have no genuine two-$\rho$ terms.

## 6.1 Group I, Row II (I.II$^{\text{gen}}$)

**Input** (with c.c.):

$$\begin{aligned}\frac{d^3N}{d^3k} &= -\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\log\frac{\vee}{\Lambda}\int e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2(x-w)^2}\Big[-\frac{2}{\epsilon}-2\gamma-\log\frac{(x-w)^2\mu_{\overline{MS}}^2}{4}\Big]\\ &\quad\times\big[U^{ab'}(x')\rho^{b'}(x')-U^{ab'}(w')\rho^{b'}(x')\big]\big[U^{ab}(w)\rho^{b}(x)-U^{ab}(x)\rho^{b}(x)\big]+\text{c.c.}\\ &= -\pi N_c\,\mathcal{N}\int e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,B(x,w)\,\big[U(x')-U(w')\big]^{ab'}\big[U(w)-U(x)\big]^{ab}\rho^{b'}(x')\rho^{b}(x)+\text{c.c.}\end{aligned} \tag{6.1}$$

**Fix F6.** The bracket $B(x,w)$ is the $z$ integral of the soft-gluon kernel. In dimensional regularization ($d=2-2\epsilon$, $\frac{1}{\bar\epsilon}=\frac{1}{\epsilon}-\gamma+\log 4\pi$), using $2X\cdot Y = X^2+Y^2-(X-Y)^2$ and dropping scaleless integrals,

$$\int d^{2-2\epsilon}z\,\mu^{2\epsilon}\,K(a,b,z) = \pi\Big[\frac{1}{\bar\epsilon}-\log\big(4\pi^2\mu^2(a-b)^2\big)\Big], \qquad \int d^{2-2\epsilon}z\,\mu^{2\epsilon}\,K(a,a,z)=0. \tag{6.2}$$

The bracket must be $B(x,w)=\frac{1}{\pi}\int_z\big[K(x,w,z)-K(w,w,z)\big]=\frac{1}{\bar\epsilon}-\log(4\pi^2\mu^2(x-w)^2)$. The written bracket has the same $(x-w)^2$ dependence, but a different pole and constant. We use the integrand: $B(x,w)\to\frac{1}{\pi}[K(x,w,z)-K(w,w,z)]$. So

$$-N_c\,\mathcal{N}\int e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\big[K(x,w,z)-K(w,w,z)\big]\big[U(x')-U(w')\big]^{ab'}\big[U(w)-U(x)\big]^{ab}\rho^{b'}(x')\rho^{b}(x)+\text{c.c.} \tag{6.3}$$

**Entries.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_4$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w,z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_4$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_4$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_4$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w,z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_5$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w',w',z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_5$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w',w',z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_5$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w',w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_5$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w',w',z)$ | $+N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w,z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $+N_c$ | $U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w',z)$ | $-N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |

## 6.2 Group I, Row V ($\mathrm{I.V^{gen}}$)

**Input.** Part 2a has $\log(\vee/\Lambda)$ written. Part 2b has the $p^+$ integral still to do:

$$\begin{aligned}\frac{d^3N_{2b}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int e^{-ik\cdot(w'-z)}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\int_{x,y,z,x'}\frac{1}{p^+(y-x)^2+(k^+-p^+)(y-z)^2}\\ \times\frac{(x'-w')^i(z-x)^m}{(x'-w')^2(z-x)^2}\Big[\cdots-\frac{p^+}{k^+-p^+}X^{im}_{\rm up}+\cdots-\frac{k^+-p^+}{p^+}X^{im}_{\rm low}+\ldots\Big]\\ \times\big[U^{ab'}(x')-U^{ab'}(w')\big]N_c\big[U^{af}(y)-U^{af}(x)\big]\rho^{b'}(x')\rho^f(y),\end{aligned}\tag{6.4}$$

with

$$X^{im}_{\rm low}=\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2},\qquad X^{im}_{\rm up}=\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}.\tag{6.5}$$

The dots are the other four terms of the bracket. They have the factors $\frac{p^+(k^+-p^+)}{k^{+2}}$, $\frac{k^+-p^+}{k^+}$, $\frac{p^+}{k^+}$ and 1, are finite at both ends, and give no logarithm.

**The two $\Lambda$ ends.** The range $\Lambda<p^+<k^+-\Lambda$ has two $\Lambda$ ends and no $\vee$ end (Sec. 3.2).

- At $p^+\to\Lambda$ the denominator becomes $k^+(y-z)^2$ and the phase becomes 1:

$$\int_\Lambda\frac{dp^+}{2\pi}\,\frac{-\frac{k^+}{p^+}X_{\rm low}}{k^+(y-z)^2}=-\frac{1}{2\pi}\frac{X_{\rm low}}{(y-z)^2}\log\frac{k^+}{\Lambda}+\ldots\tag{6.6}$$

- At $p^+\to k^+-\Lambda$ the denominator becomes $k^+(y-x)^2$ and the phase becomes $e^{-ik\cdot(z-x)}$; with $e^{-ik\cdot(w'-z)}$ this gives $e^{-ik\cdot(w'-x)}$:

$$\int^{k^+-\Lambda}\frac{dp^+}{2\pi}\,\frac{-\frac{k^+}{k^+-p^+}X_{\rm up}}{k^+(y-x)^2}=-\frac{1}{2\pi}\frac{X_{\rm up}}{(y-x)^2}\log\frac{k^+}{\Lambda}+\ldots\tag{6.7}$$

With $\frac{g^4}{8\pi^4}\cdot\frac{1}{2\pi}=\frac{g^4}{16\pi^5}$ and $\log(k^+/\Lambda)=L_\Lambda-L_k$, the $L_\Lambda$ part of 2b has exactly the two kernels of part 2a. Adding part 2a, whose colour factor is $-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z)+N_cU^{af}(x)$, the $N_cU^{af}(x)$ terms cancel:

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\mathrm{I.V},\,2\rho} = -\mathcal{N}\int\Big\{&e^{-ik\cdot(w'-z)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\Big[\mathcal{K}(z-x)\cdot\mathcal{K}(x-z)+\tfrac12\mathcal{K}(z-x)\cdot\mathcal{K}(y-x)\Big]\\ &+e^{-ik\cdot(w'-x)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-x)\Big[\mathcal{K}(z-x)\cdot\mathcal{K}(x-z)-\tfrac12\mathcal{K}(z-x)\cdot\mathcal{K}(y-z)\Big]\Big\}\\ &\times\big[U^{ab'}(x')-U^{ab'}(w')\big]\Big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z)+N_cU^{af}(y)\Big]\rho^{b'}(x')\rho^f(y)\;+\;\text{c.c.}\end{aligned}\tag{6.8}$$

In the first line the measured gluon is at $z$ and the unobserved gluon at $x$; in the second it is the other way round. In common variables we rename accordingly ($z\to w$, $x\to z$ in the first line; $x\to w$ in the second) and use $\mathcal{K}(z-x)\cdot\mathcal{K}(x-z)=-1/(x-z)^2$. The $L_k$ part is $-1$ times the $L_\Lambda$ part, so $P_{\mathrm{I.V}}=0$.

**Entries.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_4$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$ | $-2$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)$ |
| $S_4$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$ | $+2$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)$ |
| $S_4$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$ | $+2N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_4$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$ | $-2N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_5$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$ | $-2$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)$ |
| $S_5$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$ | $+2$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)$ |
| $S_5$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$ | $+2N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_5$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$ | $-2N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)$ |
| $S_7$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $+N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)$ |
| $S_{10}$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)$ |

## 6.3 Group II, Row I (II.I$^{\text{gen}}$)

**Input** (no c.c.). The kernel of the notes simplifies:

$$\frac{(x-z)^2-\frac{(x-w)^2(x'-z)^2}{(x'-w')^2}}{(x'-w')^2(x-z)^2(x-w)^2-(x-w)^4(x'-z)^2}=\frac{1}{(x-w)^2(x'-w')^2}. \tag{6.9}$$

With $\frac{g^4}{8\pi^5}=2\cdot\frac{g^4}{16\pi^5}$ the row is

$$\begin{aligned}&2\mathcal{N}\int e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\Big[K(w,w',z)-\tfrac{1}{2}K(x',w,z)-\tfrac{1}{2}K(x,w',z)+\tfrac{1}{4}K(x,x',z)\Big]\\&\times\Big[f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}f^{acd}U^{ac'}(w')U^{bc}(z)U^{de}(x)\rho^{d'}(x')\rho^{e}(x)\\&\quad-f^{ac'd'}f^{cbd}U^{ac}(w)U^{bc'}(z)U^{e'd'}(x')\rho^{e'}(x')\rho^{d}(x)+N_cU^{e'd}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x)\Big].\end{aligned} \tag{6.10}$$

**Entries.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)$ | $+2$ | $f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)$ | $-2$ | $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)$ | $-2$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)$ | $+2N_c$ | $U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{db}(z)\,U^{cl}(x')\,U^{ae}(w)\,\rho^l(x')\,\rho^h(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{db}(z)\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |

## 6.4 Group II, Row IV (II.IV$^{\text{gen}}$)

**Input** (no c.c.; $y\to w$, $y'\to w'$):

$$2\pi N_c\mathcal{N}\int e^{-ik(w'-w)}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,B'(w,w')\,\big[U^{e'c}(w')-U^{e'c}(x')\big]\big[U^{ec}(w)-U^{ec}(x)\big]\rho^{e'}(x')\rho^{e}(x), \tag{6.11}$$

with $B'(w,w')=\frac{1}{\bar\epsilon}-\log(4\pi^2\mu^2(w-w')^2)=\frac{1}{\pi}\int_z K(w,w',z)$, as written (Eq. (6.2)). For the pointwise comparison we use the integrand with the same integral,

$$B'(w,w')\to\frac{1}{\pi}\Big[K(w,w',z)-\tfrac{1}{2}K(w,w,z)-\tfrac{1}{2}K(w',w',z)\Big]=-\frac{1}{2\pi}\Big[K(w,w,z)-2K(w,w',z)+K(w',w',z)\Big], \tag{6.12}$$

the dipole kernel, which is free of infrared divergences. The $u \to \infty$ end of this row is $\mathfrak{P}_{\mathrm{II.IV}}$ of Sec. 3.3.

**Entries.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_4$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w,z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_4$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_4$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_4$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w,z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_5$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w',w',z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_5$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w',w',z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_5$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w',w',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_5$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w',w',z)$ | $+N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w',z)$ | $+2N_c$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w',z)$ | $+2N_c$ | $U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w',z)$ | $-2N_c$ | $U^{ab}(w')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w',z)$ | $-2N_c$ | $U^{ab}(x')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)$ |

## 6.5 Group III, Row III, part 2 (III.III$^{\mathrm{gen}}$)

**Input** (with c.c.; $y' \to w'$):

$$2\mathcal{N}\int e^{-ik(w'-w)}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\Big[-K(w,w',z)+\tfrac{1}{2}K(x,w',z)\Big]$$
$$\times\Big[f^{c'd'a}U^{e'c'}(w')U^{d'b}(z)-f^{c'd'a}U^{bd'}(z)U^{c'e'}(x')\Big]\rho^{e'}(x')\Big[f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\Big]+\text{c.c.}\quad(6.13)$$

**Fix F5.** The overall sign of this term is reversed. With the sign as written, Group III Row III disagrees with the independent leading-log computation of the row (earlier note, row-by-row table); with the reversed sign it agrees. The entries below have the reversed sign.

**Entries.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w',z)$ | $-2$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w',z)$ | $-2$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w',z)$ | $+2$ | $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w',z)$ | $+2$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w',z)$ | $-4N_c$ | $U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w',z)$ | $+2N_c$ | $U^{ab}(w')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)$ |
| $S_6$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(w,w',z)$ | $+2N_c$ | $U^{ab}(x')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)$ |
| $S_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,w',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{db}(z)\,U^{cl}(x')\,U^{ae}(w)\,\rho^l(x')\,\rho^h(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w,z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')$ |
| $S_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',w,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{db}(z)\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |

# 7 Two-$\rho$ terms from reordering the four-$\rho$ terms

Reordering a four-$\rho$ term gives its symmetrized four-$\rho$ part, three-$\rho$ terms and two-$\rho$ terms. The two-$\rho$ terms come in two pieces:

- the two-$\rho$ terms produced directly by the reordering: the lists "Two $\rho$ contribution" of `4rho.tex` (Sec. 7.1);
- the two-$\rho$ terms of the three-$\rho$ remainders A1, A2, B1, B2, after one more reordering step (Sec. 7.2).

The complete two-$\rho$ content of Eq. (4rho) (with its c.c.) was computed by an exact (Weyl) decomposition of every four-$\rho$ product. It equals the sum of the two pieces exactly, in all kernel structures, once fixes F1 and F2 are made. Both pieces are needed, and neither may be counted twice.

## 7.1 The lists of `4rho.tex` ($4\rho$)

**Input.** The two-$\rho$ terms of `4rho.tex` in the present notation (kernel $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)$ is the LO kernel of Eq. (4rho) with $u\to y$; units of $\mathcal{N}$):

| term | coeff. | kernel | colour |
|---|---|---|---|
| first part, 1 | $+2\,N_c$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',x',z)$ | $U^{ca}(x')\,U^{cb}(w)\,\rho^a(x')\,\rho^b(y)$ |
| first part, 2 | $-2\,N_c$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',x',z)$ | $U^{ca}(x')\,U^{cb}(y)\,\rho^a(x')\,\rho^b(y)$ |
| first part, 3 | $+2\,N_c$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $U^{ca}(y)\,U^{cb}(w')\,\rho^b(x')\,\rho^a(y)$ |
| first part, 4 | $-2\,N_c$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $U^{ca}(y)\,U^{cb}(x')\,\rho^b(x')\,\rho^a(y)$ |
| first part, 5 | $+\frac{3}{2}\,N_c$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$ | $U^{bc}(x')\,U^{ac}(y)\,\rho^b(x')\,\rho^a(y)$ |
| first part, 6 (drop) | $-8\,N_c$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,K(y,x',z)$ | $U^{bc}(x')\,U^{ca}(y)\,\rho^b(x')\,\rho^a(y)$ |
| $dN^{(1)}$, 1 | $-1$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$ | $f^{abc}\,f^{def}\,U^{eb}(w)\,U^{b'f}(x')\,U^{da}(y)\,\rho^{b'}(x')\,\rho^c(y)$ |
| $dN^{(1)}$, 2 (as printed) | $+3$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{deb'}\,U^{ab'}(y)\,U^{db'}(w')\,U^{eb}(z)\,\rho^{b'}(x')\,\rho^c(y)$ |
| $dN^{(1)}$, 2 (corrected) | $+3$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{deg}\,U^{ag}(y)\,U^{db'}(w')\,U^{eb}(z)\,\rho^{b'}(x')\,\rho^c(y)$ |
| $dN^{(1)}$, 3 | $+\frac{1}{2}$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$ | $f^{abc}\,f^{ade}\,U^{db'}(w')\,U^{b'b}(w)\,\rho^e(x')\,\rho^c(y)$ |
| $dN^{(1)}$, 4 | $-1$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$ | $f^{abc}\,f^{ade}\,U^{b'b}(w)\,U^{b'd}(x')\,\rho^e(x')\,\rho^c(y)$ |
| $dN^{(1)}$, 5 | $+2$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$ | $f^{abc}\,f^{ade}\,U^{b'b}(y)\,U^{b'd}(x')\,\rho^e(x')\,\rho^c(y)$ |
| $dN^{(1)}$, 6 | $+1$ | $\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(x',y,z)$ | $f^{abc}\,f^{dae}\,U^{b'b}(w')\,U^{b'd}(x')\,\rho^e(x')\,\rho^c(y)$ |
| $dN^{(1)}$, 7 | $-3$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{af}(z)\,U^{bd'}(w)\,U^{df}(y)\,U^{b'd'}(w')\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(1)}$, 8 | $+2$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{af}(z)\,U^{bd'}(y)\,U^{df}(y)\,U^{b'd'}(w')\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(1)}$, 9 | $-3$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{af}(w)\,U^{bd'}(y)\,U^{dd'}(z)\,U^{b'f}(w')\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(1)}$, 10 | $+3$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{ad'}(z)\,U^{df}(w)\,U^{fb'}(w')\,U^{d'b}(y)\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(1)}$, 11 | $+1$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{fd}(w)\,U^{fb'}(x')\,U^{d'a}(z)\,U^{d'b}(y)\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(1)}$, 12 | $+\frac{1}{2}$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$ | $f^{abc}\,f^{d'fl}\,U^{ad'}(z)\,U^{fb}(w)\,U^{b'l}(x')\,\rho^{b'}(x')\,\rho^c(y)$ |
| $dN^{(1)}$, 13 | $+\frac{1}{2}$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$ | $f^{abc}\,f^{d'fl}\,U^{ad'}(z)\,U^{bf}(w')\,U^{b'l}(y)\,\rho^c(x')\,\rho^{b'}(y)$ |
| $dN^{(1)}$, 14 | $-1$ | $\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(x',y,z)$ | $f^{abc}\,f^{d'fl}\,U^{fb}(y)\,U^{b'l}(x')\,U^{d'a}(z)\,\rho^{b'}(x')\,\rho^c(y)$ |
| $dN^{(1)}$, 15 | $-1$ | $\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{fb}(y)\,U^{fb'}(w)\,U^{d'a}(z)\,U^{d'd}(y)\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(1)}$, 16 | $+1$ | $\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{fb}(y)\,U^{fb'}(x')\,U^{d'a}(z)\,U^{d'd}(y)\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(1)}$, 17 | $+2$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{ad'}(z)\,U^{bf}(y)\,U^{dd'}(y)\,U^{b'f}(x')\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(1)}$, 18 | $+1$ | $\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{fd}(y)\,U^{fb'}(w)\,U^{d'a}(z)\,U^{d'b}(y)\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(1)}$, 19 | $-2$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{fd}(y)\,U^{fb'}(w')\,U^{d'a}(z)\,U^{d'b}(y)\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(1)}$, 20 | $-2$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{fd}(y)\,U^{fb'}(x')\,U^{d'a}(z)\,U^{d'b}(y)\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(1)}$, 21 | $-1$ | $\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{fd}(y)\,U^{fb'}(x')\,U^{d'a}(z)\,U^{d'b}(y)\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(2)}$, 1 | $+1$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{d'a}(w)\,U^{d'b'}(x')\,U^{fb}(z)\,U^{fd}(y)\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(2)}$, 2 | $+1$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{cde}\,U^{d'b}(w)\,U^{d'b'}(x')\,U^{fa}(y)\,U^{fd}(z)\,\rho^{b'}(x')\,\rho^e(y)$ |
| $dN^{(2)}$, 3 | $-1$ | $\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(x',y,z)$ | $f^{fd'l}\,f^{deb'}\,U^{df}(z)\,U^{cl}(y)\,U^{d'e}(x')\,\rho^{b'}(x')\,\rho^c(y)$ |
| $dN^{(2)}$, 4 | $+1$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$ | $f^{abc}\,f^{deb'}\,U^{ad}(z)\,U^{bb'}(y)\,U^{ef}(x')\,\rho^f(x')\,\rho^c(y)$ |

Fix F1: as printed, the $-8N_c$ term has nothing to cancel against; it is dropped. Fix F2: term 2 of $dN^{(1)}$ cannot be evaluated (the index $b'$ occurs four times); only the assignment shown in blue reproduces the exact decomposition.

**Entries** (in common variables, collected):

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $+2N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $-2$ | $f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(x)\,U^{lm}(w')\,U^{lb}(x)\,\rho^m(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $-2$ | $f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(x)\,U^{lm}(w')\,U^{le}(x)\,\rho^m(x')\,\rho^c(x)$ |

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $+3$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $-2N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $-2$ | $f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(x)\,U^{lm}(x')\,U^{lb}(x)\,\rho^m(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $-2$ | $f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(x)\,U^{lm}(x')\,U^{le}(x)\,\rho^m(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $+3$ | $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $-3$ | $f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^m(x')\,\rho^e(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $+3$ | $f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^m(x')\,\rho^e(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^l(x')\,\rho^e(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',x',z)$ | $+2N_c$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',x',z)$ | $+1$ | $f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(x')\,U^{lb}(x')\,U^{lm}(w)\,\rho^c(x')\,\rho^m(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',x',z)$ | $+1$ | $f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(x')\,U^{le}(x')\,U^{lm}(w)\,\rho^c(x')\,\rho^m(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',x',z)$ | $-2N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',x',z)$ | $-1$ | $f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(x')\,U^{lb}(x')\,U^{lm}(x)\,\rho^c(x')\,\rho^m(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x',x',z)$ | $-1$ | $f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(x')\,U^{le}(x')\,U^{lm}(x)\,\rho^c(x')\,\rho^m(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $+2$ | $f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $+\frac{3N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |

## 7.2 The remainders A1, A2, B1, B2 (A,B)

**A1: Two Wilson lines, first term.** In the notes this term reads $-\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{\log(\vee/\Lambda)}{2\pi^2}\frac{1}{k^+}\int(\dots)$, i.e. its prefactor is $-i\mathcal{N}$. It is the complete remainder of reordering the four-$\rho$ terms, so no c.c. is added.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\rm A1} \;=\; -\,i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_1(x',x,y)\,\Big\{ \\ (1)\;+\;f^{abc}\,U^{de}(w')\,U^{eb}(w)\,\rho^d(x')\,\rho^a(x)\,\rho^c(y) \\ (2)\;-\;f^{abc}\,U^{eb}(w)\,U^{ed}(x')\,\rho^d(x')\,\rho^a(x)\,\rho^c(y)\Big\}\end{aligned} \tag{7.1}$$

with $K_1(x',x,y)=\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$. Reordering: Eq. (4.1) with $A$, $B$, $C$ the charges at $x'$, $x$, $y$.

**A2: Two Wilson lines, second term.** Same prefactor $-i\mathcal{N}$ and no c.c.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\rm A2} \;=\; -\,i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_2(x',x,y)\,\Big\{ \\ (1)\;+\;f^{abc}\,U^{eb}(w')\,U^{ed}(y)\,\rho^c(x')\,\rho^a(x)\,\rho^d(y) \\ (2)\;-\;f^{abc}\,U^{be}(w')\,U^{ed}(w)\,\rho^c(x')\,\rho^a(x)\,\rho^d(y)\Big\}\end{aligned} \tag{7.2}$$

with $K_2(x',x,y)=\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$. Reordering: Eq. (4.1) with $A$, $B$, $C$ the charges at $x'$, $x$, $y$.

**B1: Three Wilson lines, kernel $K_1$.** In the notes the three-Wilson-line remainder has prefactor $+\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{\log(\vee/\Lambda)}{2\pi^2}\frac{1}{k^+}=+i\mathcal{N}$. Its first and fourth terms share the kernel $K_1$ and are treated here; no c.c. is added.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\rm B1} = i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_1(x',x,y)\,\Big\{ \\ (1)\;-\;f^{abc}\,U^{db}(w')\,U^{ec}(y)\,U^{af}(x)\,\rho^d(x')\,\rho^f(x)\,\rho^e(y) \\ (2)\;+\;f^{abc}\,U^{ec}(y)\,U^{bd}(x')\,U^{af}(x)\,\rho^d(x')\,\rho^f(x)\,\rho^e(y)\Big\}\end{aligned} \tag{7.3}$$

with $K_1(x',x,y)=\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$. Reordering: Eq. (4.1) with $A$, $B$, $C$ the charges at $x'$, $x$, $y$.

B2: **Three Wilson lines, kernel $K_3$.** The second and third terms of the three-Wilson-line remainder share the kernel $K_3$; prefactor $+i\mathcal{N}$, no c.c.

$$\left.\frac{d^3N}{d^3k}\right|_{\rm B2} = i\,\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_3(x',x,y)\,\Big\{ \\ (1)\;+\;f^{abc}\,U^{ec}(x')\,U^{db}(w)\,U^{af}(y)\,\rho^e(x')\,\rho^d(x)\,\rho^f(y) \\ (2)\;-\;f^{abc}\,U^{ec}(x')\,U^{bd}(x)\,U^{af}(y)\,\rho^e(x')\,\rho^d(x)\,\rho^f(y)\Big\} \tag{7.4}$$

with $K_3(x',x,y)=\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z)$. Reordering: Eq. (4.1) with $A$, $B$, $C$ the charges at $x'$, $x$, $y$.

**Two-$\rho$ terms.**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$ |
| $S_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_{11}$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$ |

# 8 The sum of all $\log(\vee/\Lambda)$ terms: JIMWLK

## 8.1 JIMWLK⊗LO

With $\mathcal{N}_0 L_\Lambda\,\mathcal{J}$ the JIMWLK evolution of the LO cross section over $\delta Y = L_\Lambda$ ($\mathcal{N}=\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\cdot\frac{\alpha_s}{2\pi^2}\delta Y$ is the LO prefactor times $\frac{\alpha_s}{2\pi^2}\delta Y$):

$$\mathcal{N}\,\mathcal{J} = -\mathcal{N}\int e^{-ik\cdot(w'-w)}\int_{x,x'}\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\;\tilde{H}\Big\{\rho^{b'}(x')\big[U(x')-U(w')\big]^{ab'}\big[U(x)-U(w)\big]^{ab}\rho^b(x)\Big\}, \\ \tilde{H} = \int_{u,v,z}K(u,v,z)\Big\{J_R^h(u)J_R^h(v)+J_L^m(u)J_L^m(v)-U^{mh}(z)\big[J_R^h(u)J_L^m(v)+J_L^m(u)J_R^h(v)\big]\Big\}, \tag{8.1}$$

with $J_R^h(u)\,U(p)=\delta^{(2)}(u-p)\,U(p)T^h$, $J_L^m(u)=U^{mg}(u)J_R^g(u)$, $(T^h)_{bc}=-if^{hbc}$. This is your $H_{\rm JIMWLK}$ of Eq. (HLO), with $d/dY=-H$. The derivatives act on the Wilson lines only: the soft gluon $z$ is emitted from the charges (through $U(x)$, $U(x')$) and from the measured gluon (through $U(w)$, $U(w')$). Carrying out the derivatives and simplifying the colour algebra gives

$$\mathcal{J} = \int e^{-ik\cdot(w'-w)}\int_{x,x',z}\Big\{ \\ +\underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)}_{S_1}\Big[ \\ +2N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\ -2N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\ +2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x) \\ -2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)\Big]$$

$$
\begin{aligned}
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)}_{S_2}\Big[\\
&\quad + 2N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad - 2N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)\Big]\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)}_{S_3}\Big[\\
&\quad + 2N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&\quad + 2\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\Big]\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)}_{S_4}\Big[\\
&\quad - 2N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad + 2N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)\Big]\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)}_{S_5}\Big[\\
&\quad - 2N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad + 2N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)\Big]\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)}_{S_6}\Big[\\
&\quad + 2N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad + 2\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)\Big]\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)}_{S_8}\Big[\\
&\quad - 2N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&\quad - 2\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\Big]\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)}_{S_9}\Big[\\
&\quad - 2N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad - 2\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)\Big]\Big\}
\end{aligned}
\tag{8.2}
$$

(The corrected Appendix F of `main.tex` equals Eq. (8.1); see the two-$\rho$ note.)

## 8.2 The comparison

For every kernel structure and every independent colour structure, the coefficients contributed by the sources add up to the coefficient in $\mathcal{J}$ (last column). Coefficients are in units of $\mathcal{N}$. Within a kernel structure the colour structures of the entry tables are expressed in a basis of independent ones (App. A); the coefficients are exact polynomials in $N_c$, fixed from SU(3) and SU(4). In total there are 154 independent colour structures in 13 kernel structures (164 distinct colour structures before the reduction). All of them match.

**Structure $S_1$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | I.III$^{3\rho}$: $\frac{N_c}{2}$; I.IV$^{3\rho}$: $-\frac{N_c}{2}$; III.I$^{3\rho}$: $\frac{N_c}{2}$; A,B: $-\frac{N_c}{2}$; $4\rho$: $2N_c$ | $2N_c$ | $2N_c$ ✓ |
| $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | I.III$^{3\rho}$: $-\frac{N_c}{2}$; I.IV$^{3\rho}$: $\frac{N_c}{2}$; II.I$^{3\rho}$: $\frac{N_c}{2}$; III.I$^{3\rho}$: $-\frac{N_c}{2}$; III.II$^{3\rho}$: $-\frac{N_c}{2}$; A,B: $\frac{N_c}{2}$; $4\rho$: $-2N_c$ | $-2N_c$ | $-2N_c$ ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')$ $\rho^c(x)$ | I.III$^{3\rho}$: $\frac{1}{2}$; II.I$^{3\rho}$: $\frac{1}{2}$; $4\rho$: 1 | 2 | 2 ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')$ $\rho^c(x)$ | I.III$^{3\rho}$: $-\frac{1}{2}$; III.II$^{3\rho}$: $-\frac{1}{2}$; $4\rho$: $-1$ | $-2$ | $-2$ ✓ |
| $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | III.II$^{3\rho}$: $-\frac{N_c}{2}$; A,B: $\frac{N_c}{2}$ | 0 | 0 ✓ |
| $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | II.I$^{3\rho}$: $\frac{N_c}{2}$; A,B: $-\frac{N_c}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)$ $\rho^l(x')\,\rho^e(x)$ | I.IV$^{3\rho}$: $\frac{1}{2}$; II.I$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$; III.II$^{3\rho}$: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)$ $U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ | I.IV$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$ | 0 | 0 ✓ |

**Structure $S_2$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | I.III$^{3\rho}$: $\frac{N_c}{2}$; I.IV$^{3\rho}$: $-\frac{N_c}{2}$; III.I$^{3\rho}$: $\frac{N_c}{2}$; A,B: $-\frac{N_c}{2}$; $4\rho$: $2N_c$ | $2N_c$ | $2N_c$ ✓ |
| $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | I.III$^{3\rho}$: $-\frac{N_c}{2}$; I.IV$^{3\rho}$: $\frac{N_c}{2}$; II.I$^{3\rho}$: $\frac{N_c}{2}$; III.I$^{3\rho}$: $-\frac{N_c}{2}$; III.II$^{3\rho}$: $-\frac{N_c}{2}$; A,B: $\frac{N_c}{2}$; $4\rho$: $-2N_c$ | $-2N_c$ | $-2N_c$ ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(x')\,\rho^c(x')$ $\rho^l(x)$ | I.III$^{3\rho}$: $\frac{1}{2}$; II.I$^{3\rho}$: $\frac{1}{2}$; $4\rho$: 1 | 2 | 2 ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(x')\,\rho^c(x')$ $\rho^l(x)$ | I.III$^{3\rho}$: $-\frac{1}{2}$; III.II$^{3\rho}$: $-\frac{1}{2}$; $4\rho$: $-1$ | $-2$ | $-2$ ✓ |
| $U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$ | III.II$^{3\rho}$: $-\frac{N_c}{2}$; A,B: $\frac{N_c}{2}$ | 0 | 0 ✓ |
| $U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$ | II.I$^{3\rho}$: $\frac{N_c}{2}$; A,B: $-\frac{N_c}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)$ $U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ | I.IV$^{3\rho}$: $-\frac{1}{2}$; II.I$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$; III.II$^{3\rho}$: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)$ $U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ | I.IV$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$ | 0 | 0 ✓ |

**Structure $S_3$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | I.II$^{3\rho}$: $-2N_c$; I.III$^{3\rho}$: $N_c$; I.IV$^{3\rho}$: $N_c$; III.I$^{3\rho}$: $-N_c$; III.II$^{3\rho}$: $N_c$; II.I$^{\text{gen}}$: $\frac{N_c}{2}$; $4\rho$: $\frac{3N_c}{2}$ | $2N_c$ | $2N_c$ ✓ |
| $f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$ | $4\rho$: 2 | 2 | 2 ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^l(x')$ $\rho^c(x)$ | I.III$^{3\rho}$: $-\frac{1}{2}$; III.II$^{3\rho}$: $-\frac{1}{2}$; $4\rho$: $-1$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^c(x')$ $\rho^l(x)$ | I.III$^{3\rho}$: $-\frac{1}{2}$; III.II$^{3\rho}$: $-\frac{1}{2}$; $4\rho$: $-1$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ | III.II$^{3\rho}$: $\frac{1}{2}$; A,B: $\frac{1}{2}$; $4\rho$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)$ | III.II$^{3\rho}$: $-\frac{1}{2}$; A,B: $-\frac{1}{2}$; $4\rho$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ | II.I$^{\text{gen}}$: $\frac{1}{2}$; A,B: $-1$; $4\rho$: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)$ $\rho^h(x')$ | I.II$^{3\rho}$: 1; I.III$^{3\rho}$: $-\frac{1}{2}$; I.IV$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$; A,B: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')$ $\rho^h(x)$ | I.II$^{3\rho}$: $-1$; I.III$^{3\rho}$: $\frac{1}{2}$; I.IV$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$; A,B: $-\frac{1}{2}$; $4\rho$: 1 | 0 | 0 ✓ |

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ | I.III$^{3\rho}$: $-\frac{1}{2}$; I.IV$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$; III.II$^{3\rho}$: $\frac{1}{2}$; II.I$^{\text{gen}}$: $\frac{1}{2}$; $4\rho$: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ | I.III$^{3\rho}$: $\frac{1}{2}$; I.IV$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$; III.II$^{3\rho}$: $-\frac{1}{2}$; II.I$^{\text{gen}}$: $-\frac{1}{2}$; $4\rho$: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)$ | I.IV$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)$ | I.IV$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$ | 0 | 0 ✓ |

**Structure $S_4$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | I.II$^{\text{gen}}$: $-N_c$; II.IV$^{\text{gen}}$: $-N_c$ | $-2N_c$ | $-2N_c$ ✓ |
| $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | I.II$^{\text{gen}}$: $N_c$; II.IV$^{\text{gen}}$: $N_c$ | $2N_c$ | $2N_c$ ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)$ | I.V$^{\text{gen}}$: $-2$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)$ | I.V$^{\text{gen}}$: 2 | 2 | 2 ✓ |
| $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | I.II$^{\text{gen}}$: $-N_c$; I.V$^{\text{gen}}$: $2N_c$; II.IV$^{\text{gen}}$: $-N_c$ | 0 | 0 ✓ |
| $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | I.II$^{\text{gen}}$: $N_c$; I.V$^{\text{gen}}$: $-2N_c$; II.IV$^{\text{gen}}$: $N_c$ | 0 | 0 ✓ |

**Structure $S_5$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | I.II$^{\text{gen}}$: $-N_c$; II.IV$^{\text{gen}}$: $-N_c$ | $-2N_c$ | $-2N_c$ ✓ |
| $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | I.II$^{\text{gen}}$: $N_c$; II.IV$^{\text{gen}}$: $N_c$ | $2N_c$ | $2N_c$ ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)$ | I.V$^{\text{gen}}$: $-2$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)$ | I.V$^{\text{gen}}$: 2 | 2 | 2 ✓ |
| $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | I.II$^{\text{gen}}$: $-N_c$; I.V$^{\text{gen}}$: $2N_c$; II.IV$^{\text{gen}}$: $-N_c$ | 0 | 0 ✓ |
| $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | I.II$^{\text{gen}}$: $N_c$; I.V$^{\text{gen}}$: $-2N_c$; II.IV$^{\text{gen}}$: $N_c$ | 0 | 0 ✓ |

**Structure $S_6$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)$

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | II.IV$^{\text{gen}}$: $2N_c$ | $2N_c$ | $2N_c$ ✓ |
| $f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ | II.I$^{\text{gen}}$: 2 | 2 | 2 ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)$ | III.III$^{\text{gen}}$: $-2$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)$ | III.III$^{\text{gen}}$: $-2$ | $-2$ | $-2$ ✓ |
| $U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$ | II.I$^{\text{gen}}$: $2N_c$; II.IV$^{\text{gen}}$: $2N_c$; III.III$^{\text{gen}}$: $-4N_c$ | 0 | 0 ✓ |
| $U^{ab}(w')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)$ | II.IV$^{\text{gen}}$: $-2N_c$; III.III$^{\text{gen}}$: $2N_c$ | 0 | 0 ✓ |
| $U^{ab}(x')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)$ | II.IV$^{\text{gen}}$: $-2N_c$; III.III$^{\text{gen}}$: $2N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ | II.I$^{\text{gen}}$: $-2$; III.III$^{\text{gen}}$: 2 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)$ | II.I$^{\text{gen}}$: $-2$; III.III$^{\text{gen}}$: 2 | 0 | 0 ✓ |

**Structure $S_7$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$

JIMWLK⊗LO has no term with this kernel: the terms cancel among themselves.

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | III.II$^{3\rho}$: $N_c$; I.II$^{\text{gen}}$: $-N_c$ | 0 | 0 ✓ |
| $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | $\mathcal{D}$: $N_c$; I.IV$^{3\rho}$: $-N_c$; II.I$^{3\rho}$: $-N_c$; III.I$^{3\rho}$: $N_c$; III.II$^{3\rho}$: $N_c$; III.III$^{3\rho}$: $N_c$; III.V$^{3\rho}$: $-N_c$; III.VI$^{3\rho}$: $-N_c$; I.II$^{\text{gen}}$: $N_c$; I.V$^{\text{gen}}$: $-N_c$ | 0 | 0 ✓ |
| $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | II.I$^{3\rho}$: $-N_c$; I.II$^{\text{gen}}$: $N_c$ | 0 | 0 ✓ |

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | $\mathcal{D}$: $-N_c$; I.IV$^{3\rho}$: $N_c$; III.I$^{3\rho}$: $-N_c$; III.V$^{3\rho}$: $N_c$; I.II$^{\text{gen}}$: $-N_c$; I.V$^{\text{gen}}$: $N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x)$ | $\mathcal{D}$: $-1$; III.III$^{3\rho}$: $-1$; III.V$^{3\rho}$: $1$; III.VI$^{3\rho}$: $1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x)$ | $\mathcal{D}$: $1$; III.V$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)$ | III.VI$^{3\rho}$: $-1$; I.V$^{\text{gen}}$: $1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)$ | III.III$^{3\rho}$: $1$; I.V$^{\text{gen}}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x)$ | II.I$^{3\rho}$: $-1$; III.III$^{3\rho}$: $1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x)$ | III.II$^{3\rho}$: $1$; III.VI$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ | I.IV$^{3\rho}$: $-1$; II.I$^{3\rho}$: $-1$; III.I$^{3\rho}$: $1$; III.II$^{3\rho}$: $1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ | I.IV$^{3\rho}$: $1$; III.I$^{3\rho}$: $-1$ | 0 | 0 ✓ |

**Structure $S_8$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | $\mathcal{D}$: $-N_c$; III.V$^{3\rho}$: $N_c$; III.VI$^{3\rho}$: $-N_c$; III.III$^{\text{gen}}$: $-N_c$ | $-2N_c$ | $-2N_c$ ✓ |
| $f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$ | I.II$^{3\rho}$: $-1$; III.II$^{3\rho}$: $-1$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)$ | I.V$^{3\rho}$: $1$; III.VI$^{3\rho}$: $1$ | 2 | 2 ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ | I.IV$^{3\rho}$: $1$; III.I$^{3\rho}$: $-1$; III.II$^{3\rho}$: $1$; II.I$^{\text{gen}}$: $1$ | 2 | 2 ✓ |
| $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | I.II$^{3\rho}$: $N_c$; $\mathcal{D}$: $N_c$; I.IV$^{3\rho}$: $-N_c$; I.V$^{3\rho}$: $-N_c$; III.I$^{3\rho}$: $N_c$; III.II$^{3\rho}$: $-N_c$; III.V$^{3\rho}$: $-N_c$; III.VI$^{3\rho}$: $N_c$; II.I$^{\text{gen}}$: $-N_c$; III.III$^{\text{gen}}$: $N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)$ | I.II$^{3\rho}$: $-1$; II.I$^{\text{gen}}$: $1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$ | I.II$^{3\rho}$: $1$; $\mathcal{D}$: $1$; I.IV$^{3\rho}$: $-1$; I.V$^{3\rho}$: $-1$; III.I$^{3\rho}$: $1$; III.V$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(w')\,\rho^l(x')\,\rho^h(x)$ | $\mathcal{D}$: $-1$; III.V$^{3\rho}$: $1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)$ | I.V$^{3\rho}$: $-1$; III.III$^{\text{gen}}$: $1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{db}(z)\,U^{cl}(x')\,U^{ae}(w)\,\rho^l(x')\,\rho^h(x)$ | II.I$^{\text{gen}}$: $-1$; III.III$^{\text{gen}}$: $1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)$ | III.II$^{3\rho}$: $-1$; III.VI$^{3\rho}$: $1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)$ | I.IV$^{3\rho}$: $-1$; III.I$^{3\rho}$: $1$ | 0 | 0 ✓ |

**Structure $S_9$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | $\mathcal{D}$: $-N_c$; III.V$^{3\rho}$: $N_c$; III.VI$^{3\rho}$: $-N_c$; III.III$^{\text{gen}}$: $-N_c$ | $-2N_c$ | $-2N_c$ ✓ |
| $f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ | I.II$^{3\rho}$: $-1$; III.II$^{3\rho}$: $-1$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$ | I.IV$^{3\rho}$: $1$; III.I$^{3\rho}$: $-1$; III.II$^{3\rho}$: $1$; II.I$^{\text{gen}}$: $1$ | 2 | 2 ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)$ | I.V$^{3\rho}$: $1$; III.VI$^{3\rho}$: $1$ | 2 | 2 ✓ |
| $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ | I.II$^{3\rho}$: $N_c$; $\mathcal{D}$: $N_c$; I.IV$^{3\rho}$: $-N_c$; I.V$^{3\rho}$: $-N_c$; III.I$^{3\rho}$: $N_c$; III.II$^{3\rho}$: $-N_c$; III.V$^{3\rho}$: $-N_c$; III.VI$^{3\rho}$: $N_c$; II.I$^{\text{gen}}$: $-N_c$; III.III$^{\text{gen}}$: $N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ | I.II$^{3\rho}$: $1$; II.I$^{\text{gen}}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$ | I.II$^{3\rho}$: $-1$; $\mathcal{D}$: $-1$; I.IV$^{3\rho}$: $1$; I.V$^{3\rho}$: $1$; III.I$^{3\rho}$: $-1$; III.V$^{3\rho}$: $1$ | 0 | 0 ✓ |

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(w)\,\rho^h(x')\,\rho^l(x)$ | $\mathcal{D}$: 1; III.V$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')$ | I.V$^{3\rho}$: 1; III.III$^{\text{gen}}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{db}(z)\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ | II.I$^{\text{gen}}$: $-1$; III.III$^{\text{gen}}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ | III.II$^{3\rho}$: 1; III.VI$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)$ | I.IV$^{3\rho}$: 1; III.I$^{3\rho}$: $-1$ | 0 | 0 ✓ |

**Structure $S_{10}$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$

JIMWLK⊗LO has no term with this kernel: the terms cancel among themselves.

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$ | III.II$^{3\rho}$: $N_c$; I.II$^{\text{gen}}$: $-N_c$ | 0 | 0 ✓ |
| $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | $\mathcal{D}$: $N_c$; I.IV$^{3\rho}$: $-N_c$; II.I$^{3\rho}$: $-N_c$; III.I$^{3\rho}$: $N_c$; III.II$^{3\rho}$: $N_c$; III.III$^{3\rho}$: $N_c$; III.V$^{3\rho}$: $-N_c$; III.VI$^{3\rho}$: $-N_c$; I.II$^{\text{gen}}$: $N_c$; I.V$^{\text{gen}}$: $-N_c$ | 0 | 0 ✓ |
| $U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$ | II.I$^{3\rho}$: $-N_c$; I.II$^{\text{gen}}$: $N_c$ | 0 | 0 ✓ |
| $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | $\mathcal{D}$: $-N_c$; I.IV$^{3\rho}$: $N_c$; III.I$^{3\rho}$: $-N_c$; III.V$^{3\rho}$: $N_c$; I.II$^{\text{gen}}$: $-N_c$; I.V$^{\text{gen}}$: $N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x)$ | $\mathcal{D}$: 1; III.III$^{3\rho}$: 1; III.V$^{3\rho}$: $-1$; III.VI$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x)$ | $\mathcal{D}$: $-1$; III.V$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)$ | III.VI$^{3\rho}$: 1; I.V$^{\text{gen}}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)$ | III.III$^{3\rho}$: $-1$; I.V$^{\text{gen}}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ce}(x')\,U^{al}(w)\,\rho^l(x)\,\rho^h(x')$ | II.I$^{3\rho}$: 1; III.III$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')$ | III.II$^{3\rho}$: $-1$; III.VI$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ | I.IV$^{3\rho}$: 1; II.I$^{3\rho}$: 1; III.I$^{3\rho}$: $-1$; III.II$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ | I.IV$^{3\rho}$: $-1$; III.I$^{3\rho}$: 1 | 0 | 0 ✓ |

**Structure $S_{11}$:** $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$

JIMWLK⊗LO has no term with this kernel: the terms cancel among themselves.

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | I.II$^{3\rho}$: $2N_c$; I.III$^{3\rho}$: $-N_c$; I.IV$^{3\rho}$: $-N_c$; II.I$^{3\rho}$: $-N_c$; III.I$^{3\rho}$: $N_c$; III.II$^{3\rho}$: $N_c$; A,B: $-N_c$ | 0 | 0 ✓ |
| $U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | I.III$^{3\rho}$: $N_c$; III.II$^{3\rho}$: $-N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$ | III.II$^{3\rho}$: $-\frac{1}{2}$; A,B: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')$ | III.II$^{3\rho}$: $\frac{1}{2}$; A,B: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$ | II.I$^{3\rho}$: $\frac{1}{2}$; A,B: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')$ | II.I$^{3\rho}$: $-\frac{1}{2}$; A,B: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$ | I.II$^{3\rho}$: 1; I.III$^{3\rho}$: $-\frac{1}{2}$; I.IV$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$; A,B: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$ | I.II$^{3\rho}$: $-1$; I.III$^{3\rho}$: $\frac{1}{2}$; I.IV$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$; A,B: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)$ | I.III$^{3\rho}$: $-\frac{1}{2}$; II.I$^{3\rho}$: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x')$ | I.III$^{3\rho}$: $\frac{1}{2}$; II.I$^{3\rho}$: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ | I.IV$^{3\rho}$: $\frac{1}{2}$; II.I$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$; III.II$^{3\rho}$: $-\frac{1}{2}$ | 0 | 0 ✓ |

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)$ $\rho^m(x)\,\rho^e(x')$ | I.IV$^{3\rho}$: $-\frac{1}{2}$; II.I$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$; III.II$^{3\rho}$: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)$ $\rho^m(x)\,\rho^e(x')$ | I.IV$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)$ $\rho^m(x)\,\rho^e(x')$ | I.IV$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$ | 0 | 0 ✓ |

**Structure $S_{12}$:** $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$

JIMWLK⊗LO has no term with this kernel: the terms cancel among themselves.

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | I.II$^{3\rho}$: $-N_c$; $\mathcal{D}$: $-N_c$; I.IV$^{3\rho}$: $N_c$; I.V$^{3\rho}$: $N_c$; II.I$^{3\rho}$: $N_c$; III.I$^{3\rho}$: $-N_c$; III.II$^{3\rho}$: $-N_c$; III.III$^{3\rho}$: $-N_c$; III.V$^{3\rho}$: $N_c$; III.VI$^{3\rho}$: $N_c$ | 0 | 0 ✓ |
| $U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$ | III.II$^{3\rho}$: $N_c$; III.VI$^{3\rho}$: $-N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$ | I.II$^{3\rho}$: $-1$; III.II$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$ | I.II$^{3\rho}$: 1; II.I$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')$ $\rho^l(x)$ | I.II$^{3\rho}$: $-1$; $\mathcal{D}$: $-1$; I.IV$^{3\rho}$: 1; I.V$^{3\rho}$: 1; III.I$^{3\rho}$: $-1$; III.V$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)$ $\rho^h(x')$ | $\mathcal{D}$: $-1$; III.III$^{3\rho}$: $-1$; III.V$^{3\rho}$: 1; III.VI$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)$ $\rho^h(x')$ | $\mathcal{D}$: 1; III.V$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')$ $\rho^l(x)$ | I.V$^{3\rho}$: $-1$; III.VI$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')$ $\rho^l(x)$ | I.V$^{3\rho}$: 1; III.III$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x')\,\rho^l(x)$ $\rho^h(x')$ | II.I$^{3\rho}$: $-1$; III.III$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)$ $\rho^m(x)\,\rho^e(x')$ | I.IV$^{3\rho}$: $-1$; II.I$^{3\rho}$: $-1$; III.I$^{3\rho}$: 1; III.II$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)$ $\rho^m(x)\,\rho^e(x')$ | I.IV$^{3\rho}$: 1; III.I$^{3\rho}$: $-1$ | 0 | 0 ✓ |

**Structure $S_{13}$:** $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$

JIMWLK⊗LO has no term with this kernel: the terms cancel among themselves.

| colour structure | contributions (source: coefficient) | sum | JIMWLK |
|---|---|---|---|
| $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ | I.II$^{3\rho}$: $-N_c$; $\mathcal{D}$: $-N_c$; I.IV$^{3\rho}$: $N_c$; I.V$^{3\rho}$: $N_c$; II.I$^{3\rho}$: $N_c$; III.I$^{3\rho}$: $-N_c$; III.II$^{3\rho}$: $-N_c$; III.III$^{3\rho}$: $-N_c$; III.V$^{3\rho}$: $N_c$; III.VI$^{3\rho}$: $N_c$ | 0 | 0 ✓ |
| $U^{ab}(z)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)$ | III.II$^{3\rho}$: $N_c$; III.VI$^{3\rho}$: $-N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')$ | I.II$^{3\rho}$: 1; III.II$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')$ | I.II$^{3\rho}$: $-1$; II.I$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)$ $\rho^h(x')$ | I.II$^{3\rho}$: 1; $\mathcal{D}$: 1; I.IV$^{3\rho}$: $-1$; I.V$^{3\rho}$: $-1$; III.I$^{3\rho}$: 1; III.V$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)$ $\rho^h(x')$ | $\mathcal{D}$: 1; III.III$^{3\rho}$: 1; III.V$^{3\rho}$: $-1$; III.VI$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)$ $\rho^h(x')$ | $\mathcal{D}$: $-1$; III.V$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)$ $\rho^h(x')$ | I.V$^{3\rho}$: 1; III.VI$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)$ $\rho^h(x')$ | I.V$^{3\rho}$: $-1$; III.III$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ce}(x')\,U^{ad}(w)\,\rho^h(x')$ $\rho^l(x)$ | II.I$^{3\rho}$: 1; III.III$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)$ $\rho^m(x)\,\rho^e(x')$ | I.IV$^{3\rho}$: 1; II.I$^{3\rho}$: 1; III.I$^{3\rho}$: $-1$; III.II$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)$ $\rho^m(x)\,\rho^e(x')$ | I.IV$^{3\rho}$: $-1$; III.I$^{3\rho}$: 1 | 0 | 0 ✓ |

# 9 The coefficient of $\log\vee$: BFKL

## 9.1 $P_{\rm row}$ for every row

The coefficient of $\log\vee$ of a row is its integrand at the $\vee$ end, $P_{\rm row} = A_{\rm row} + B_{\rm row}$ (Sec. 3.2). The coefficients of $\log(\vee/k^+)$, $B_{\rm row}$, were extracted row by row from the notes in the $\log(\vee/k^+)$ note (explicit through $\log(\Lambda/k^+)$, $\log\frac{\vee}{\Lambda+k^+}$, $\log(\frac{\vee}{k^+}-1)$, or hidden in plus-prescription integrals and in logarithms of transverse distances). For the two-$\rho$ terms:

- *four-$\rho$ terms* ($4\rho$, A,B): $P = A$. The entries are those of Sec. 7.
- *Group I Row V* (I.V$^{3\rho}$, I.V$^{\rm gen}$), *the genuine Group I Row II term* (I.II$^{\rm gen}$) and $\mathcal{D}$: $P = 0$.
- *Group I Row II* (I.II$^{3\rho}$): only the $\mathfrak{F}_\alpha$ part survives, with the sign reversed (fix F4).
- *all other rows*: the integrand at the $\vee$ end, listed below.

The entry tables below use the kernel structures $S'_n$ of the $\log\vee$ terms (Table 4); coefficients in units of $\mathcal{N}_0\log\vee$.

**I.II$^{3\rho}$ (fix F4: $-\mathfrak{F}_\alpha$)**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+2$ | $f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(x')\,\rho^b(x)\,\rho^e(x')$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |

**I.III$^{3\rho}$ (with fix P1)**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S'_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S'_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S'_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S'_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S'_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S'_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S'_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$ |
| $S'_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-N_c$ | $U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-N_c$ | $U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$ |

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-N_c$ | $U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |

**I.IV**$^{3\rho}$

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_6'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_6'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_6'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_6'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x)$ |

**II.I**$^{3\rho}$

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^{c}(x)\,\rho^{b}(x')$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{b}(x')\,\rho^{e}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)$ |

**III.I$^{3\rho}$**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S'_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$ |
| $S'_1$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$ |
| $S'_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$ |
| $S'_2$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S'_4$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)$ |

**III.II**$^{3\rho}$

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(x')\,\rho^b(x)\,\rho^e(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+N_c$ | $U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_6'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_6'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_6'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_6'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-2$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+N_c$ | $U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-2N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ |
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x')$ |
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$ |
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-2$ | $f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+N_c$ | $U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-2N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x)$ |

**III.III**$^{3\rho}$

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_6'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)$ |
| $S_6'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_6'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_6'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x')$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)$ |
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+N_c$ | $U^{ab}(z)\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')$ |
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$ |
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_8'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |

**III.V$^{3\rho}$**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+N_c$ | $U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $+N_c$ | $U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+N_c$ | $U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |

**III.VI$^{3\rho}$**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(z)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $+2$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_6$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+2$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(z)\,\rho^{b}(x)\,\rho^{e}(x')$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-2N_c$ | $U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+2$ | $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+2N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-N_c$ | $U^{ab}(z)\,U^{ac}(x)\,\rho^{c}(x)\,\rho^{b}(x')$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-2$ | $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)$ |
| $S'_8$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+2$ | $f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(x)\,\rho^{b}(x')\,\rho^{e}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-2N_c$ | $U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-2$ | $f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+2N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |

**II.I$^{\text{gen}}$**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S'_3$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_5$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{b}(x)\,\rho^{e}(x')$ |
| $S'_5$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{b}(x')\,\rho^{e}(x)$ |
| $S'_5$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x')\,\rho^{c}(x)\,\rho^{l}(x')$ |
| $S'_5$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)$ |
| $S'_5$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S'_5$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_5$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+2N_c$ | $U^{ab}(x)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')$ |
| $S'_7$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{b}(x')\,\rho^{e}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)$ |
| $S'_9$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)$ |

**II.IV$^{\rm gen}$ (this is $\mathfrak{P}_{\rm II.IV}$ of Sec. 3.3)**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+2N_c$ | $\rho^a(x)\,\rho^a(x')$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+2N_c$ | $U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-2N_c$ | $U^{ab}(z)\,U^{cb}(x')\,\rho^a(x)\,\rho^c(x')$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-2N_c$ | $U^{ab}(z)\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)$ |

**III.III$^{\rm gen}$ (with fix F5)**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x)\,\rho^e(x')$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x')\,\rho^e(x)$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x')\,\rho^c(x)\,\rho^l(x')$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x)\,\rho^l(x)\,\rho^h(x')$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-4N_c$ | $U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+2N_c$ | $U^{ab}(z)\,U^{cb}(x')\,\rho^a(x)\,\rho^c(x')$ |
| $S_5'$ | $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+2N_c$ | $U^{ab}(z)\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x')$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $+N_c$ | $U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_7'$ | $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+N_c$ | $U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_9'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |

**4$\rho$ ($P=A$)**

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+2N_c$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-2N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-3$ | $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+3$ | $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+2$ | $f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-3$ | $f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^m(x')\,\rho^e(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+3$ | $f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^m(x')\,\rho^e(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+2$ | $f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^m(x')\,\rho^e(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-2$ | $f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-2$ | $f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^l(x')\,\rho^e(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+2N_c$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-2N_c$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+1$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-1$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+2$ | $f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+1$ | $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{3N_c}{2}$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$ |

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)$ |

**A,B** ($P=A$)

| S | kernel | coeff. | colour structure |
|---|---|---|---|
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $-\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_1'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $-\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ |
| $S_2'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $+\frac{N_c}{2}$ | $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-1$ | $f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')$ |
| $S_3'$ | $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(x')\,\rho^b(x)\,\rho^e(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-N_c$ | $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $+\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)$ |
| $S_4'$ | $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $-\frac{1}{2}$ | $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')$ |

The sources $\mathcal{D}$, I.V$^{3\rho}$, I.II$^{\rm gen}$ and I.V$^{\rm gen}$ have no entries here ($P=0$).

## 9.2 BFKL⊗LO

Let $O(x',x)=\rho^{b'}(x')\,M^{b'b}(x',x)\,\rho^b(x)$ with $M^{b'b}(p',p)=\big[U(w')-U(p')\big]^{ab'}\big[U(w)-U(p)\big]^{ab}$. The LO cross section is $\propto\int\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\langle O(x',x)\rangle$. A gluon harder than the measured one is added to the projectile: its charge $j^a(z)$ adds to $\rho^a$, and the valence charges get the virtual corrections. With $\mathcal{N}_0 L\,\mathcal{B}$ the BFKL evolution of the LO cross section over a rapidity interval $L$:

$$
\begin{aligned}
\mathcal{B}=2\int_{x,x',u,v,z} & e^{-ik\cdot(w'-w)}\,K(u,v,z)\,\Big\{\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\Big[\rho^h(u)\rho^h(v)\,O(x',x)-2\rho^h(u)\,O(x',x)\,\rho^h(v)+O(x',x)\,\rho^h(u)\rho^h(v)\Big]\\
&+\mathcal{K}(z-w')\cdot\mathcal{K}(z-w)\;\rho^{e'}(u)\rho^e(v)\,(T^{b'}T^b)_{e'e}\,M^{b'b}(z,z)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\;\rho^{e'}(u)\rho^{b'}(x')\rho^e(v)\,(T^b)_{e'e}\,M^{b'b}(x',z)\\
&+\mathcal{K}(z-w')\cdot\mathcal{K}(x-w)\;\rho^{e'}(u)\rho^b(x)\rho^e(v)\,(T^{b'})_{e'e}\,M^{b'b}(z,x)\Big\}.
\end{aligned}
\tag{9.1}
$$

- The first line is the virtual correction (the double commutator $-\frac{1}{2}[\rho^h(u),[\rho^h(v),O]]$).
- The second line has both measured gluons emitted by the gluon at $z$.
- The last two lines have one measured gluon emitted by a valence charge and the other by the gluon at $z$.

In the colour-singlet projection this is Eq. (LOBFKL) of your Appendix E (checked to machine precision in the $\log(\vee/k^+)$ note). After reordering the charges and simplifying the colour algebra, its three- and four-$\rho$ parts vanish, and

$$
\begin{aligned}
\mathcal{B}=\int e^{-ik\cdot(w'-w)}\int_{x,x',z}\Big\{ & \\
&+\underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)}_{S_1'}\;\Big[\\
&\quad -N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad +N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&\quad +N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad -N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\Big]
\end{aligned}
$$

$$
\begin{aligned}
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\; K(x',x',z)}_{S_2'} \Big[ \\
&\qquad - N_c\, U^{ab}(w')\, U^{ac}(w)\, \rho^b(x')\, \rho^c(x) \\
&\qquad + N_c\, U^{ab}(w')\, U^{ac}(x)\, \rho^b(x')\, \rho^c(x) \\
&\qquad + N_c\, U^{ab}(x')\, U^{ac}(w)\, \rho^b(x')\, \rho^c(x) \\
&\qquad - N_c\, U^{ab}(x')\, U^{ac}(x)\, \rho^b(x')\, \rho^c(x) \Big] \\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\; K(x,x',z)}_{S_3'} \Big[ \\
&\qquad + 2\, f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{hd}(w)\, \rho^c(x')\, \rho^e(x) \\
&\qquad - 2\, f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{hd}(x)\, \rho^c(x')\, \rho^e(x) \\
&\qquad - 2\, f^{abc}\, f^{dbe}\, U^{ha}(x')\, U^{hd}(w)\, \rho^c(x')\, \rho^e(x) \\
&\qquad + 2\, f^{abc}\, f^{dbe}\, U^{ha}(x')\, U^{hd}(x)\, \rho^c(x')\, \rho^e(x) \Big] \\
&+ \underbrace{\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\; K(x,x',z)}_{S_5'} \Big[ \\
&\qquad + 2N_c\, \rho^a(x)\, \rho^a(x') \\
&\qquad - f^{abc}\, f^{dce}\, U^{ha}(w')\, U^{hd}(w)\, \rho^b(x)\, \rho^e(x') \\
&\qquad - f^{abc}\, f^{dce}\, U^{ha}(w')\, U^{hd}(w)\, \rho^b(x')\, \rho^e(x) \\
&\qquad + f^{abc}\, f^{dce}\, U^{ha}(w')\, U^{hd}(z)\, \rho^b(x)\, \rho^e(x') \\
&\qquad + f^{abc}\, f^{dce}\, U^{ha}(w')\, U^{hd}(z)\, \rho^b(x')\, \rho^e(x) \\
&\qquad + f^{abc}\, f^{dce}\, U^{ha}(z)\, U^{hd}(w)\, \rho^b(x)\, \rho^e(x') \\
&\qquad + f^{abc}\, f^{dce}\, U^{ha}(z)\, U^{hd}(w)\, \rho^b(x')\, \rho^e(x) \Big] \\
&+ \underbrace{\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\; K(x,x,z)}_{S_6'} \Big[ \\
&\qquad - N_c\, U^{ab}(w')\, U^{ac}(w)\, \rho^b(x')\, \rho^c(x) \\
&\qquad + N_c\, U^{ab}(w')\, U^{ac}(z)\, \rho^b(x')\, \rho^c(x) \\
&\qquad + N_c\, U^{ab}(x')\, U^{ac}(w)\, \rho^b(x')\, \rho^c(x) \\
&\qquad - N_c\, U^{ab}(x')\, U^{ac}(z)\, \rho^b(x')\, \rho^c(x) \Big] \\
&+ \underbrace{\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\; K(x,x',z)}_{S_7'} \Big[ \\
&\qquad + 2\, f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(w)\, \rho^b(x)\, \rho^e(x') \\
&\qquad - 2\, f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(z)\, \rho^b(x)\, \rho^e(x') \\
&\qquad - 2\, f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(w)\, \rho^b(x)\, \rho^e(x') \\
&\qquad + 2\, f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(z)\, \rho^b(x)\, \rho^e(x') \Big] \\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\; K(x',x',z)}_{S_8'} \Big[ \\
&\qquad - N_c\, U^{ab}(w')\, U^{ac}(w)\, \rho^c(x)\, \rho^b(x') \\
&\qquad + N_c\, U^{ab}(w')\, U^{ac}(x)\, \rho^c(x)\, \rho^b(x') \\
&\qquad + N_c\, U^{ab}(z)\, U^{ac}(w)\, \rho^c(x)\, \rho^b(x') \\
&\qquad - N_c\, U^{ab}(z)\, U^{ac}(x)\, \rho^c(x)\, \rho^b(x') \Big] \\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\; K(x,x',z)}_{S_9'} \Big[ \\
&\qquad + 2\, f^{abc}\, f^{cde}\, U^{ha}(w')\, U^{hd}(w)\, \rho^b(x')\, \rho^e(x) \\
&\qquad - 2\, f^{abc}\, f^{cde}\, U^{ha}(w')\, U^{hd}(x)\, \rho^b(x')\, \rho^e(x) \\
&\qquad - 2\, f^{abc}\, f^{cde}\, U^{ha}(z)\, U^{hd}(w)\, \rho^b(x')\, \rho^e(x)
\end{aligned}
$$

$$+\,2\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x)\Big]\Big\} \tag{9.2}$$

## 9.3 The comparison

As in Sec. 8.2, now for the coefficient of $\log\vee$ against $\mathcal{B}$: 114 independent colour structures in 9 kernel structures (123 distinct colour structures before the reduction). All of them match.

**Structure $S_1'$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$

| colour structure | contributions (source: coefficient) | sum | BFKL |
|---|---|---|---|
| $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | II.I$^{3\rho}$: $-\frac{N_c}{2}$; A,B: $-\frac{N_c}{2}$ | $-N_c$ | $-N_c$ ✓ |
| $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | I.III$^{3\rho}$: $-\frac{N_c}{2}$; I.IV$^{3\rho}$: $\frac{N_c}{2}$; III.I$^{3\rho}$: $-\frac{N_c}{2}$; A,B: $-\frac{N_c}{2}$; $4\rho$: $2N_c$ | $N_c$ | $N_c$ ✓ |
| $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | III.II$^{3\rho}$: $\frac{N_c}{2}$; A,B: $\frac{N_c}{2}$ | $N_c$ | $N_c$ ✓ |
| $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | I.III$^{3\rho}$: $\frac{N_c}{2}$; I.IV$^{3\rho}$: $-\frac{N_c}{2}$; II.I$^{3\rho}$: $-\frac{N_c}{2}$; III.I$^{3\rho}$: $\frac{N_c}{2}$; III.II$^{3\rho}$: $\frac{N_c}{2}$; A,B: $\frac{N_c}{2}$; $4\rho$: $-2N_c$ | $-N_c$ | $-N_c$ ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')$ $\rho^h(x)$ | I.III$^{3\rho}$: $-\frac{1}{2}$; III.II$^{3\rho}$: $-\frac{1}{2}$; $4\rho$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')$ $\rho^h(x)$ | I.III$^{3\rho}$: $\frac{1}{2}$; II.I$^{3\rho}$: $\frac{1}{2}$; $4\rho$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)$ $\rho^l(x')\,\rho^e(x)$ | I.IV$^{3\rho}$: $-\frac{1}{2}$; II.I$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$; III.II$^{3\rho}$: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)$ $U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ | I.IV$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$ | 0 | 0 ✓ |

**Structure $S_2'$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$

| colour structure | contributions (source: coefficient) | sum | BFKL |
|---|---|---|---|
| $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | II.I$^{3\rho}$: $-\frac{N_c}{2}$; A,B: $-\frac{N_c}{2}$ | $-N_c$ | $-N_c$ ✓ |
| $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | III.II$^{3\rho}$: $\frac{N_c}{2}$; A,B: $\frac{N_c}{2}$ | $N_c$ | $N_c$ ✓ |
| $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | I.III$^{3\rho}$: $-\frac{N_c}{2}$; I.IV$^{3\rho}$: $\frac{N_c}{2}$; III.I$^{3\rho}$: $-\frac{N_c}{2}$; A,B: $-\frac{N_c}{2}$; $4\rho$: $2N_c$ | $N_c$ | $N_c$ ✓ |
| $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | I.III$^{3\rho}$: $\frac{N_c}{2}$; I.IV$^{3\rho}$: $-\frac{N_c}{2}$; II.I$^{3\rho}$: $-\frac{N_c}{2}$; III.I$^{3\rho}$: $\frac{N_c}{2}$; III.II$^{3\rho}$: $\frac{N_c}{2}$; A,B: $\frac{N_c}{2}$; $4\rho$: $-2N_c$ | $-N_c$ | $-N_c$ ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')$ $\rho^l(x)$ | I.III$^{3\rho}$: $\frac{1}{2}$; III.II$^{3\rho}$: $\frac{1}{2}$; $4\rho$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')$ $\rho^l(x)$ | I.III$^{3\rho}$: $-\frac{1}{2}$; II.I$^{3\rho}$: $-\frac{1}{2}$; $4\rho$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)$ $U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ | I.IV$^{3\rho}$: $\frac{1}{2}$; II.I$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$; III.II$^{3\rho}$: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)$ $U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ | I.IV$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$ | 0 | 0 ✓ |

**Structure $S_3'$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$

| colour structure | contributions (source: coefficient) | sum | BFKL |
|---|---|---|---|
| $f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ | I.II$^{3\rho}$: 2; II.I$^{\text{gen}}$: $\frac{1}{2}$; A,B: $-1$; $4\rho$: $\frac{1}{2}$ | 2 | 2 ✓ |
| $f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$ | I.II$^{3\rho}$: $-1$; III.II$^{3\rho}$: $-\frac{1}{2}$; A,B: $\frac{1}{2}$; $4\rho$: $-1$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$ | I.II$^{3\rho}$: $-1$; III.II$^{3\rho}$: $-\frac{1}{2}$; A,B: $\frac{1}{2}$; $4\rho$: $-1$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$ | $4\rho$: 2 | 2 | 2 ✓ |
| $U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$ | I.III$^{3\rho}$: $-N_c$; I.IV$^{3\rho}$: $-N_c$; III.I$^{3\rho}$: $N_c$; III.II$^{3\rho}$: $-N_c$; II.I$^{\text{gen}}$: $\frac{N_c}{2}$; $4\rho$: $\frac{3N_c}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)$ $\rho^h(x')$ | I.III$^{3\rho}$: $-\frac{1}{2}$; III.II$^{3\rho}$: $-\frac{1}{2}$; $4\rho$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')$ $\rho^h(x)$ | I.III$^{3\rho}$: $\frac{1}{2}$; III.II$^{3\rho}$: $\frac{1}{2}$; $4\rho$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)$ $\rho^h(x')$ | I.III$^{3\rho}$: $\frac{1}{2}$; I.IV$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$; III.II$^{3\rho}$: $-\frac{1}{2}$; II.I$^{\text{gen}}$: $\frac{1}{2}$; $4\rho$: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')$ $\rho^h(x)$ | I.III$^{3\rho}$: $-\frac{1}{2}$; I.IV$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$; III.II$^{3\rho}$: $\frac{1}{2}$; II.I$^{\text{gen}}$: $-\frac{1}{2}$; $4\rho$: $\frac{1}{2}$ | 0 | 0 ✓ |

| colour structure | contributions (source: coefficient) | sum | BFKL |
|---|---|---|---|
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')$ | I.III$^{3\rho}$: $-\frac{1}{2}$; I.IV$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$; A,B: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)$ | I.III$^{3\rho}$: $\frac{1}{2}$; I.IV$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$; A,B: $\frac{1}{2}$; $4\rho$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)$ | I.IV$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)$ | I.IV$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$ | 0 | 0 ✓ |

**Structure $S_4'$:** $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$

BFKL⊗LO has no term with this kernel: the terms cancel among themselves.

| colour structure | contributions (source: coefficient) | sum | BFKL |
|---|---|---|---|
| $U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | I.III$^{3\rho}$: $-N_c$; III.II$^{3\rho}$: $N_c$ | 0 | 0 ✓ |
| $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | I.III$^{3\rho}$: $N_c$; I.IV$^{3\rho}$: $N_c$; II.I$^{3\rho}$: $N_c$; III.I$^{3\rho}$: $-N_c$; III.II$^{3\rho}$: $-N_c$; A,B: $-N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hb}(w')\,U^{ha}(w)\,\rho^d(x)\,\rho^e(x')$ | I.II$^{3\rho}$: $-1$; II.I$^{3\rho}$: $\frac{1}{2}$; A,B: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x')$ | I.II$^{3\rho}$: 2; II.I$^{3\rho}$: $-1$; A,B: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(x')\,\rho^b(x)\,\rho^e(x')$ | I.II$^{3\rho}$: $-1$; III.II$^{3\rho}$: $\frac{1}{2}$; A,B: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{ha}(w)\,\rho^d(x)\,\rho^e(x')$ | I.II$^{3\rho}$: 1; III.II$^{3\rho}$: $-\frac{1}{2}$; A,B: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{ha}(x')\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x')$ | I.II$^{3\rho}$: $-1$; III.II$^{3\rho}$: $\frac{1}{2}$; A,B: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)$ | I.III$^{3\rho}$: $\frac{1}{2}$; II.I$^{3\rho}$: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x')$ | I.III$^{3\rho}$: $-\frac{1}{2}$; II.I$^{3\rho}$: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)$ | I.III$^{3\rho}$: $-\frac{1}{2}$; I.IV$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$; A,B: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')$ | I.III$^{3\rho}$: $\frac{1}{2}$; I.IV$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$; A,B: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ | I.IV$^{3\rho}$: $-\frac{1}{2}$; II.I$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$; III.II$^{3\rho}$: $\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ | I.IV$^{3\rho}$: $\frac{1}{2}$; II.I$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$; III.II$^{3\rho}$: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ | I.IV$^{3\rho}$: $\frac{1}{2}$; III.I$^{3\rho}$: $-\frac{1}{2}$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$ | I.IV$^{3\rho}$: $-\frac{1}{2}$; III.I$^{3\rho}$: $\frac{1}{2}$ | 0 | 0 ✓ |

**Structure $S_5'$:** $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$

| colour structure | contributions (source: coefficient) | sum | BFKL |
|---|---|---|---|
| $\rho^a(x)\,\rho^a(x')$ | II.IV$^{\text{gen}}$: $2N_c$ | $2N_c$ | $2N_c$ ✓ |
| $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x')$ | II.I$^{\text{gen}}$: $-1$ | $-1$ | $-1$ ✓ |
| $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)$ | II.I$^{\text{gen}}$: $-1$ | $-1$ | $-1$ ✓ |
| $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x)\,\rho^e(x')$ | III.III$^{\text{gen}}$: 1 | 1 | 1 ✓ |
| $f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x')\,\rho^e(x)$ | III.III$^{\text{gen}}$: 1 | 1 | 1 ✓ |
| $f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x')$ | III.III$^{\text{gen}}$: 1 | 1 | 1 ✓ |
| $f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)$ | III.III$^{\text{gen}}$: 1 | 1 | 1 ✓ |
| $U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$ | II.I$^{\text{gen}}$: $2N_c$; II.IV$^{\text{gen}}$: $2N_c$; III.III$^{\text{gen}}$: $-4N_c$ | 0 | 0 ✓ |
| $U^{ab}(z)\,U^{cb}(x')\,\rho^a(x)\,\rho^c(x')$ | II.IV$^{\text{gen}}$: $-2N_c$; III.III$^{\text{gen}}$: $2N_c$ | 0 | 0 ✓ |
| $U^{ab}(z)\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)$ | II.IV$^{\text{gen}}$: $-2N_c$; III.III$^{\text{gen}}$: $2N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x')\,\rho^c(x)\,\rho^l(x')$ | II.I$^{\text{gen}}$: $-1$; III.III$^{\text{gen}}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$ | II.I$^{\text{gen}}$: $-1$; III.III$^{\text{gen}}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x)\,\rho^l(x)\,\rho^h(x')$ | II.I$^{\text{gen}}$: $-1$; III.III$^{\text{gen}}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)$ | II.I$^{\text{gen}}$: $-1$; III.III$^{\text{gen}}$: 1 | 0 | 0 ✓ |

**Structure $S_6'$:** $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$

| colour structure | contributions (source: coefficient) | sum | BFKL |
|---|---|---|---|
| $U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | $\mathrm{II.I}^{3\rho}\colon -N_c$ | $-N_c$ | $-N_c$ ✓ |
| $U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)$ | $\mathrm{III.III}^{3\rho}\colon N_c$ | $N_c$ | $N_c$ ✓ |
| $U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$ | $\mathrm{III.II}^{3\rho}\colon N_c$ | $N_c$ | $N_c$ ✓ |
| $U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)$ | $\mathrm{III.VI}^{3\rho}\colon -N_c$ | $-N_c$ | $-N_c$ ✓ |
| $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | $\mathrm{I.III}^{3\rho}\colon N_c$; $\mathrm{I.IV}^{3\rho}\colon -N_c$; $\mathrm{II.I}^{3\rho}\colon -N_c$; $\mathrm{III.I}^{3\rho}\colon N_c$; $\mathrm{III.II}^{3\rho}\colon N_c$; $\mathrm{III.III}^{3\rho}\colon N_c$; $\mathrm{III.V}^{3\rho}\colon -N_c$; $\mathrm{III.VI}^{3\rho}\colon -N_c$ | 0 | 0 ✓ |
| $U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | $\mathrm{I.III}^{3\rho}\colon -N_c$; $\mathrm{I.IV}^{3\rho}\colon N_c$; $\mathrm{III.I}^{3\rho}\colon -N_c$; $\mathrm{III.V}^{3\rho}\colon N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')$ $\rho^h(x)$ | $\mathrm{I.III}^{3\rho}\colon -1$; $\mathrm{III.II}^{3\rho}\colon -1$; $\mathrm{III.III}^{3\rho}\colon -1$; $\mathrm{III.V}^{3\rho}\colon 1$; $\mathrm{III.VI}^{3\rho}\colon 2$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')$ $\rho^h(x)$ | $\mathrm{I.III}^{3\rho}\colon 1$; $\mathrm{II.I}^{3\rho}\colon 1$; $\mathrm{III.III}^{3\rho}\colon -1$; $\mathrm{III.V}^{3\rho}\colon -1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)$ $\rho^l(x')\,\rho^e(x)$ | $\mathrm{I.IV}^{3\rho}\colon -1$; $\mathrm{II.I}^{3\rho}\colon -1$; $\mathrm{III.I}^{3\rho}\colon 1$; $\mathrm{III.II}^{3\rho}\colon 1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)$ $U^{md}(x)\,\rho^l(x')\,\rho^e(x)$ | $\mathrm{I.IV}^{3\rho}\colon 1$; $\mathrm{III.I}^{3\rho}\colon -1$ | 0 | 0 ✓ |

**Structure $S_7'$:** $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$

| colour structure | contributions (source: coefficient) | sum | BFKL |
|---|---|---|---|
| $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ | $\mathrm{II.I}^{3\rho}\colon 1$; $\mathrm{II.I}^{\mathrm{gen}}\colon 1$ | 2 | 2 ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x')$ | $\mathrm{III.III}^{3\rho}\colon -1$; $\mathrm{III.III}^{\mathrm{gen}}\colon -1$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$ | $\mathrm{III.II}^{3\rho}\colon -2$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x')$ | $\mathrm{III.VI}^{3\rho}\colon 2$ | 2 | 2 ✓ |
| $U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | $\mathrm{I.III}^{3\rho}\colon -N_c$; $\mathrm{III.II}^{3\rho}\colon N_c$; $\mathrm{III.V}^{3\rho}\colon N_c$; $\mathrm{III.VI}^{3\rho}\colon -2N_c$; $\mathrm{III.III}^{\mathrm{gen}}\colon N_c$ | 0 | 0 ✓ |
| $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | $\mathrm{II.I}^{3\rho}\colon N_c$; $\mathrm{III.II}^{3\rho}\colon -2N_c$; $\mathrm{III.III}^{3\rho}\colon -N_c$; $\mathrm{III.VI}^{3\rho}\colon 2N_c$; $\mathrm{II.I}^{\mathrm{gen}}\colon N_c$; $\mathrm{III.III}^{\mathrm{gen}}\colon -N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)$ $\rho^h(x')$ | $\mathrm{I.III}^{3\rho}\colon -1$; $\mathrm{III.II}^{3\rho}\colon -1$; $\mathrm{III.III}^{3\rho}\colon -1$; $\mathrm{III.V}^{3\rho}\colon 1$; $\mathrm{III.VI}^{3\rho}\colon 2$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')$ $\rho^l(x)$ | $\mathrm{I.III}^{3\rho}\colon 1$; $\mathrm{II.I}^{3\rho}\colon -1$; $\mathrm{III.III}^{3\rho}\colon 1$; $\mathrm{III.V}^{3\rho}\colon -1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)$ $\rho^h(x')$ | $\mathrm{I.III}^{3\rho}\colon 1$; $\mathrm{III.V}^{3\rho}\colon -1$; $\mathrm{II.I}^{\mathrm{gen}}\colon 1$; $\mathrm{III.III}^{\mathrm{gen}}\colon -1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')$ $\rho^c(x)$ | $\mathrm{I.IV}^{3\rho}\colon -1$; $\mathrm{III.I}^{3\rho}\colon 1$; $\mathrm{III.II}^{3\rho}\colon -1$; $\mathrm{II.I}^{\mathrm{gen}}\colon 1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)$ $\rho^m(x)\,\rho^e(x')$ | $\mathrm{I.IV}^{3\rho}\colon -1$; $\mathrm{II.I}^{3\rho}\colon -1$; $\mathrm{III.I}^{3\rho}\colon 1$; $\mathrm{III.II}^{3\rho}\colon 1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)$ $U^{md}(x')\,\rho^h(x')\,\rho^c(x)$ | $\mathrm{I.IV}^{3\rho}\colon 1$; $\mathrm{III.I}^{3\rho}\colon -1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)$ $\rho^m(x)\,\rho^e(x')$ | $\mathrm{I.IV}^{3\rho}\colon 1$; $\mathrm{III.I}^{3\rho}\colon -1$ | 0 | 0 ✓ |

**Structure $S_8'$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$

| colour structure | contributions (source: coefficient) | sum | BFKL |
|---|---|---|---|
| $U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')$ | $\mathrm{II.I}^{3\rho}\colon -N_c$ | $-N_c$ | $-N_c$ ✓ |
| $U^{ab}(w')\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x')$ | $\mathrm{III.II}^{3\rho}\colon N_c$ | $N_c$ | $N_c$ ✓ |
| $U^{ab}(z)\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')$ | $\mathrm{III.III}^{3\rho}\colon N_c$ | $N_c$ | $N_c$ ✓ |
| $U^{ab}(z)\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x')$ | $\mathrm{III.VI}^{3\rho}\colon -N_c$ | $-N_c$ | $-N_c$ ✓ |
| $U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | $\mathrm{I.III}^{3\rho}\colon N_c$; $\mathrm{I.IV}^{3\rho}\colon -N_c$; $\mathrm{II.I}^{3\rho}\colon -N_c$; $\mathrm{III.I}^{3\rho}\colon N_c$; $\mathrm{III.II}^{3\rho}\colon N_c$; $\mathrm{III.III}^{3\rho}\colon N_c$; $\mathrm{III.V}^{3\rho}\colon -N_c$; $\mathrm{III.VI}^{3\rho}\colon -N_c$ | 0 | 0 ✓ |
| $U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$ | $\mathrm{I.III}^{3\rho}\colon -N_c$; $\mathrm{I.IV}^{3\rho}\colon N_c$; $\mathrm{III.I}^{3\rho}\colon -N_c$; $\mathrm{III.V}^{3\rho}\colon N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')$ $\rho^l(x)$ | $\mathrm{I.III}^{3\rho}\colon 1$; $\mathrm{III.II}^{3\rho}\colon 1$; $\mathrm{III.III}^{3\rho}\colon 1$; $\mathrm{III.V}^{3\rho}\colon -1$; $\mathrm{III.VI}^{3\rho}\colon -2$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')$ $\rho^l(x)$ | $\mathrm{I.III}^{3\rho}\colon -1$; $\mathrm{II.I}^{3\rho}\colon -1$; $\mathrm{III.III}^{3\rho}\colon 1$; $\mathrm{III.V}^{3\rho}\colon 1$ | 0 | 0 ✓ |

| colour structure | contributions (source: coefficient) | sum | BFKL |
|---|---|---|---|
| $f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)$ $U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ | I.IV$^{3\rho}$: 1; II.I$^{3\rho}$: 1; III.I$^{3\rho}$: $-1$; III.II$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)$ $U^{md}(x')\,\rho^e(x')\,\rho^l(x)$ | I.IV$^{3\rho}$: $-1$; III.I$^{3\rho}$: 1 | 0 | 0 ✓ |

**Structure $S_9'$:** $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$

| colour structure | contributions (source: coefficient) | sum | BFKL |
|---|---|---|---|
| $f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)$ | II.I$^{3\rho}$: 1; II.I$^{\text{gen}}$: 1 | 2 | 2 ✓ |
| $f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x)$ | III.II$^{3\rho}$: $-2$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)$ | III.III$^{3\rho}$: $-1$; III.III$^{\text{gen}}$: $-1$ | $-2$ | $-2$ ✓ |
| $f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x)$ | III.VI$^{3\rho}$: 2 | 2 | 2 ✓ |
| $U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | I.III$^{3\rho}$: $-N_c$; III.II$^{3\rho}$: $N_c$; III.V$^{3\rho}$: $N_c$; III.VI$^{3\rho}$: $-2N_c$; III.III$^{\text{gen}}$: $N_c$ | 0 | 0 ✓ |
| $U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$ | II.I$^{3\rho}$: $N_c$; III.II$^{3\rho}$: $-2N_c$; III.III$^{3\rho}$: $-N_c$; III.VI$^{3\rho}$: $2N_c$; II.I$^{\text{gen}}$: $N_c$; III.III$^{\text{gen}}$: $-N_c$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')$ $\rho^h(x)$ | I.III$^{3\rho}$: 1; III.II$^{3\rho}$: 1; III.III$^{3\rho}$: 1; III.V$^{3\rho}$: $-1$; III.VI$^{3\rho}$: $-2$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(x')$ $\rho^h(x)$ | I.III$^{3\rho}$: $-1$; II.I$^{3\rho}$: 1; III.III$^{3\rho}$: $-1$; III.V$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')$ $\rho^h(x)$ | I.III$^{3\rho}$: $-1$; III.V$^{3\rho}$: 1; II.I$^{\text{gen}}$: $-1$; III.III$^{\text{gen}}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')$ $\rho^l(x)$ | I.IV$^{3\rho}$: 1; III.I$^{3\rho}$: $-1$; III.II$^{3\rho}$: 1; II.I$^{\text{gen}}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')$ $\rho^m(x')\,\rho^e(x)$ | I.IV$^{3\rho}$: 1; II.I$^{3\rho}$: 1; III.I$^{3\rho}$: $-1$; III.II$^{3\rho}$: $-1$ | 0 | 0 ✓ |
| $f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)$ $\rho^c(x')\,\rho^h(x)$ | I.IV$^{3\rho}$: $-1$; III.I$^{3\rho}$: 1 | 0 | 0 ✓ |
| $f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')$ $\rho^m(x')\,\rho^e(x)$ | I.IV$^{3\rho}$: $-1$; III.I$^{3\rho}$: 1 | 0 | 0 ✓ |

# 10 Result

Both sums of Eq. (3.7) hold exactly:

$$\sum_{\text{sources}} A = \mathcal{J} \quad \text{(Sec. 8)}, \qquad \sum_{\text{sources}} P = \mathcal{B} \quad \text{(Sec. 9)}. \tag{10.1}$$

With Eq. (2.3) the leading-logarithmic two-$\rho$ cross section is Eq. (1.1):

$$\left.\frac{d^3N}{d^3k}\right|_{\text{LL}} = \mathcal{N}_0\Big[\log\frac{\vee}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,(\mathcal{B}-\mathcal{J})\Big] = \mathcal{N}_0\Big[\log\frac{k^+}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,\mathcal{B}\Big]. \tag{10.2}$$

Table 4 shows which kernel structure occurs where.

- Structures in which the unobserved gluon is attached to the measured gluon ($K(w,\cdot,z)$, $K(w',\cdot,z)$) occur only among the $\log(\vee/\Lambda)$ terms, never among the $\log\vee$ terms.
- Structures in which the measured gluon is emitted by the gluon at $z$ ($\mathcal{K}(w-z)$, $\mathcal{K}(w'-z)$ as emission kernels) occur only among the $\log\vee$ terms, never among the $\log(\vee/\Lambda)$ terms.
- Structures in which the unobserved gluon connects only charges occur in both, with different coefficients.

In your variables BFKL⊗LO is not the coefficient of either logarithm alone. It is the coefficient of $\log\vee$, which is the sum of the coefficients of $\log(\vee/\Lambda)$ and of $\log(\vee/k^+)$. The $\log(\vee/\Lambda)$ terms alone give JIMWLK. Taking both evolutions with $\delta Y = \log(\vee/\Lambda)$ is possible only as Eq. (1.2), i.e. with $\frac{1}{2}\delta Y$ each and a remainder $\propto \log(k^{+2}/\vee\Lambda)$.

| kernel structure | $\log(\vee/\Lambda)$ terms | $\log\vee$ terms | in $\mathcal{J}$ | in $\mathcal{B}$ |
|---|---|---|---|---|
| $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$ | $S_1$ | $S'_1$ | ✓ | ✓ |
| $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$ | $S_2$ | $S'_2$ | ✓ | ✓ |
| $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $S_3$ | $S'_3$ | ✓ | ✓ |
| $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$ | $S_4$ | $--$ | ✓ | – |
| $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$ | $S_5$ | $--$ | ✓ | – |
| $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)$ | $S_6$ | $--$ | ✓ | – |
| $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $S_7$ | $--$ | – | – |
| $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $S_8$ | $--$ | ✓ | – |
| $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$ | $S_9$ | $--$ | ✓ | – |
| $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$ | $S_{10}$ | $--$ | – | – |
| $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$ | $S_{11}$ | $S'_4$ | – | – |
| $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$ | $S_{12}$ | $--$ | – | – |
| $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$ | $S_{13}$ | $--$ | – | – |
| $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $--$ | $S'_5$ | – | ✓ |
| $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$ | $--$ | $S'_6$ | – | ✓ |
| $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$ | $--$ | $S'_7$ | – | ✓ |
| $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$ | $--$ | $S'_8$ | – | ✓ |
| $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$ | $--$ | $S'_9$ | – | ✓ |

Table 4: The kernel structures. A structure without a mark in the last two columns has terms in the rows that cancel among themselves.

| seed | $U$ | $\sum A$ | $\mathcal{J}$ | rel. diff. | $\sum P$ | $\mathcal{B}$ | rel. diff. |
|---|---|---|---|---|---|---|---|
| 3 | symmetric | −0.179957 | −0.179957 | $4.6\times10^{-16}$ | −0.011178 | −0.011178 | $2.0\times10^{-14}$ |
| 3 | general | 0.225958 | 0.286007 | $2.1\times10^{-1}$ | 0.065947 | 0.257858 | $6.7\times10^{-1}$ |
| 7 | symmetric | −515.390339 | −515.390339 | $1.1\times10^{-15}$ | −100.927484 | −100.927484 | $6.4\times10^{-15}$ |
| 7 | general | −497.394796 | −50.258248 | 4.2 | −192.799623 | 106.403972 | 2.8 |
| 11 | symmetric | 3.400970 | 3.400970 | $1.6\times10^{-15}$ | 1.899057 | 1.899057 | $7.8\times10^{-15}$ |
| 11 | general | −1.095584 | −1.142046 | $3.7\times10^{-2}$ | −1.252553 | −1.250426 | $1.7\times10^{-3}$ |
| 19 | symmetric | 4.846920 | 4.846920 | $4.0\times10^{-16}$ | −28.824389 | −28.824389 | $1.4\times10^{-14}$ |
| 19 | general | −11.591806 | −22.834296 | $4.9\times10^{-1}$ | −4.985454 | −9.938042 | $2.2\times10^{-1}$ |

Table 5: Pointwise test of both sums at random configurations (all kernels and colour factors evaluated; no integration). For symmetric Wilson lines, your convention $U^{bc}=U^{cb}$, both sums hold to machine precision. $\mathcal{J}\neq\mathcal{B}$. For general (non-symmetric) adjoint $U$ the notes do not apply, because they use $U^{bc}=U^{cb}$ throughout. This is the same as in the earlier notes.

# A Colour identities used in the basis reduction

For the kernel structures in which the simplified colour structures are linearly dependent, each dependent structure equals the following combination of basis structures (exact; checked for SU(3) and SU(4)).

## A.1 $\log(\vee/\Lambda)$ terms

**Structure $S_1$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$.**

$$f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(x)\,U^{lm}(w')\,U^{le}(x)\,\rho^m(x')\,\rho^c(x)$$
$$= f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x) - f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(x)\,U^{lm}(w')\,U^{lb}(x)\,\rho^m(x')\,\rho^c(x)$$
$$f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(x)\,U^{lm}(x')\,U^{le}(x)\,\rho^m(x')\,\rho^c(x)$$
$$= f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x) - f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(x)\,U^{lm}(x')\,U^{lb}(x)\,\rho^m(x')\,\rho^c(x)$$
$$f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^m(x')\,\rho^e(x)$$
$$= -f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) + f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^m(x')\,\rho^e(x)$$
$$f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^l(x')\,\rho^e(x)$$
$$= -f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) - f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)$$

**Structure $S_2$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$.**

$$f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(x')\,U^{le}(x')\,U^{lm}(w)\,\rho^c(x')\,\rho^m(x)$$
$$= f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x) - f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(x')\,U^{lb}(x')\,U^{lm}(w)\,\rho^c(x')\,\rho^m(x)$$
$$f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(x')\,U^{le}(x')\,U^{lm}(x)\,\rho^c(x')\,\rho^m(x)$$
$$= f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x) - f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(x')\,U^{lb}(x')\,U^{lm}(x)\,\rho^c(x')\,\rho^m(x)$$

**Structure $S_4$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$.**

$$
\begin{aligned}
&f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w)\,U^{lm}(w')\,U^{le}(w)\,\rho^{m}(x')\,\rho^{c}(x)\\
&\quad= f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(w)\,\rho^{l}(x')\,\rho^{c}(x) - f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w)\,U^{lm}(w')\,U^{lb}(w)\,\rho^{m}(x')\,\rho^{c}(x)\\
&f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w)\,U^{lm}(x')\,U^{le}(w)\,\rho^{m}(x')\,\rho^{c}(x)\\
&\quad= f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(w)\,\rho^{l}(x')\,\rho^{c}(x) - f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w)\,U^{lm}(x')\,U^{lb}(w)\,\rho^{m}(x')\,\rho^{c}(x)
\end{aligned}
$$

**Structure $S_5$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$.**

$$
\begin{aligned}
&f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w')\,U^{le}(w')\,U^{lm}(w)\,\rho^{c}(x')\,\rho^{m}(x)\\
&\quad= f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(w')\,\rho^{c}(x')\,\rho^{l}(x) - f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w')\,U^{lb}(w')\,U^{lm}(w)\,\rho^{c}(x')\,\rho^{m}(x)\\
&f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w')\,U^{le}(w')\,U^{lm}(x)\,\rho^{c}(x')\,\rho^{m}(x)\\
&\quad= f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(w')\,\rho^{c}(x')\,\rho^{l}(x) - f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w')\,U^{lb}(w')\,U^{lm}(x)\,\rho^{c}(x')\,\rho^{m}(x)
\end{aligned}
$$

## A.2 $\log\vee$ terms

**Structure $S_1'$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$.**

$$
\begin{aligned}
&f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^{m}(x')\,\rho^{e}(x)\\
&\quad= -f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x) + f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&\quad= -f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x) + f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&\quad= -f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x) + f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^{l}(x')\,\rho^{e}(x)\\
&\quad= -f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x) - f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)
\end{aligned}
$$

**Structure $S_2'$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$.**

$$
\begin{aligned}
&f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&\quad= f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x) + f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&\quad= f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x) + f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')
\end{aligned}
$$

**Structure $S_4'$, $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$.**

$$
\begin{aligned}
&f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&\quad= f^{abc}\,f^{cde}\,U^{hb}(w')\,U^{ha}(w)\,\rho^{d}(x)\,\rho^{e}(x') - f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(x')\,\rho^{b}(x)\,\rho^{e}(x')\\
&\quad= f^{abc}\,f^{cde}\,U^{hb}(w')\,U^{ha}(x')\,\rho^{d}(x)\,\rho^{e}(x') - f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(x')\,\rho^{b}(x)\,\rho^{e}(x')\\
&f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&\quad= f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{ha}(w)\,\rho^{d}(x)\,\rho^{e}(x') - f^{abc}\,f^{dce}\,U^{ha}(x')\,U^{hd}(w)\,\rho^{b}(x)\,\rho^{e}(x')
\end{aligned}
$$

# B Scripts

All lists and tables are generated by the scripts of this folder from the term definitions of `../two_rho`, `../three_rho` and `../bfkl`:

- `sets.py`: all two-$\rho$ terms, one label per source, for the coefficient of $\log(\vee/\Lambda)$ and of $\log\vee$, with the fixes;
- `checks.py`: the pointwise tests (Tables 3, 5);
- `build.py`: the exact tables by kernel structure and colour structure, and the basis reduction;
- `make_doc.py`: writes this file from `static.tex`.